ELECTRICAL ENGINEERING ASSET PORTFOLIO

Residential and Commercial Building Projects

(CAD Drawings, Schematics, Load Schedules and Technical Documentation)

Engineering Discipline: Electrical Engineering

Asset Categories:

- Electrical Power Systems
- Lighting Systems
- Extra Low Voltage (ELV) Systems
- Fire Alarm Systems
- CCTV Systems
- Load Schedules
- Consumer Unit (CU) Configurations

Software Used:

- AutoCAD
- Microsoft Excel

Document Version: Rev. 00

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Date: 6th August, 2026.

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5. Project 4 – Power, Fire Alarm & CCTV Design (Commercial Complex)
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8. Portfolio Summary

Portfolio Overview

Introduction

This portfolio presents a collection of electrical engineering design assets developed for residential and commercial building projects. The assets include electrical power layouts, lighting layouts, fire alarm systems, CCTV systems, consumer unit arrangements, schematic diagrams, and electrical load schedules.

The drawings were developed using AutoCAD, while load calculations and schedules were prepared using Microsoft Excel where applicable.

The portfolio demonstrates practical experience in producing structured engineering documentation suitable for electrical installation, contractor reference, design review, and technical evaluation.

Purpose of this Portfolio

This portfolio has been compiled to:

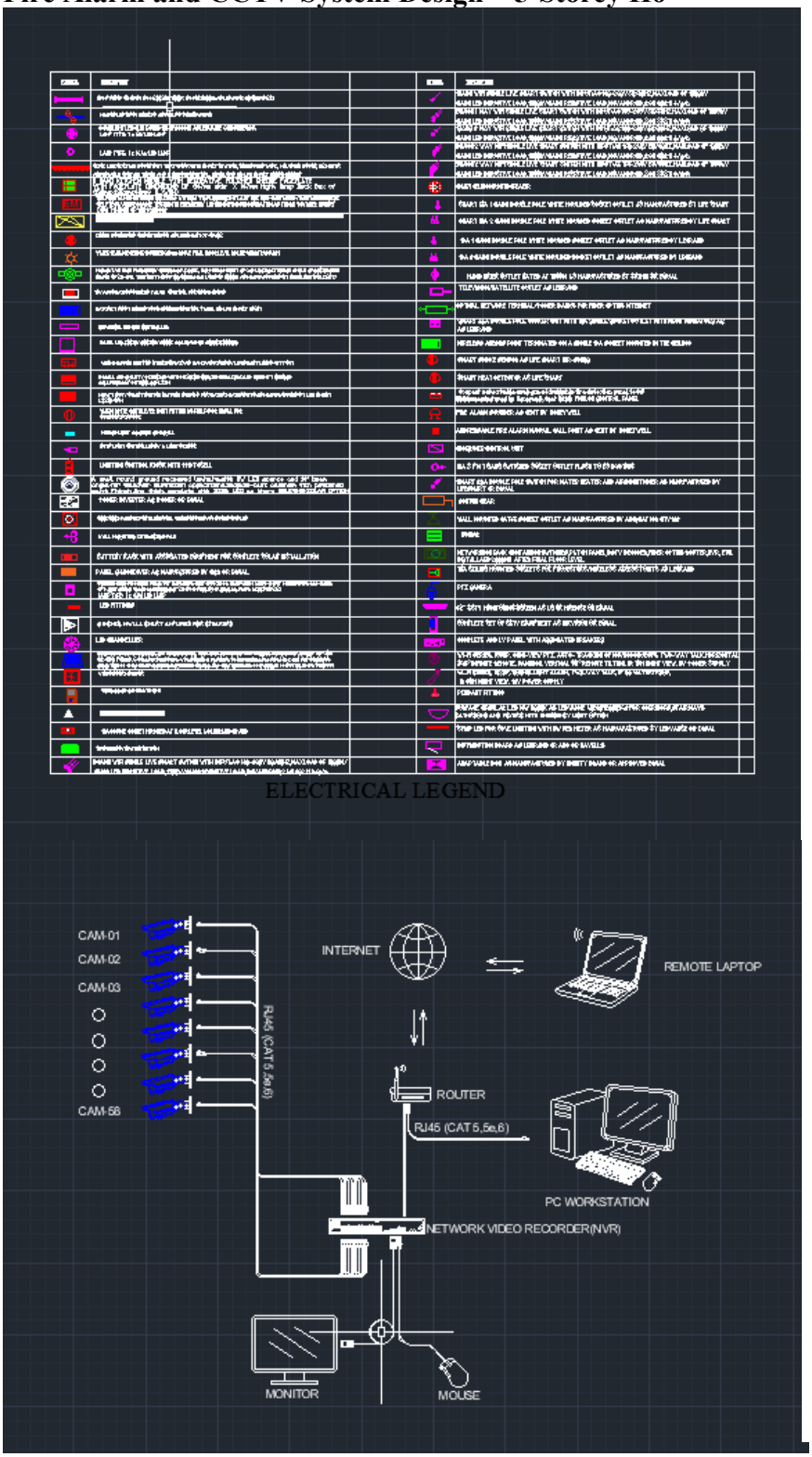
- Demonstrate competency in electrical engineering design.
- Present professionally prepared CAD drawings and supporting documentation.
- Showcase electrical power and ELV system planning.
- Provide structured engineering assets for technical review.
- Illustrate project documentation and organization practices.

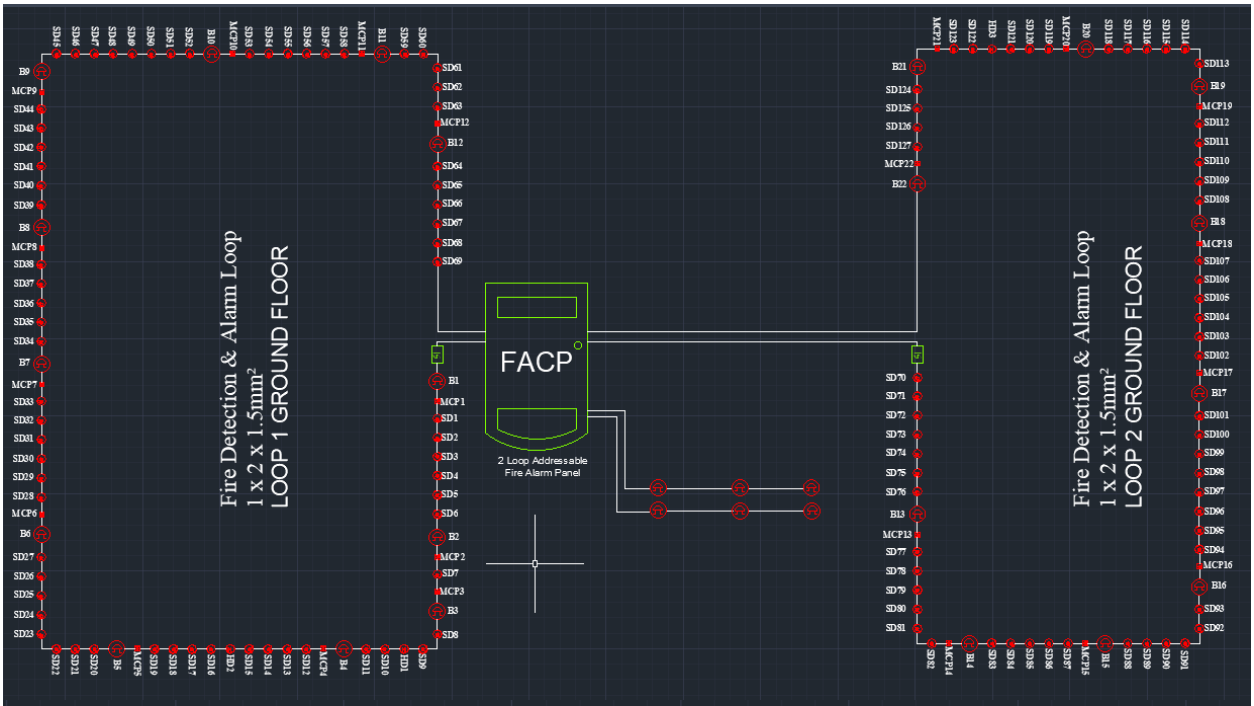
Asset Categories Included

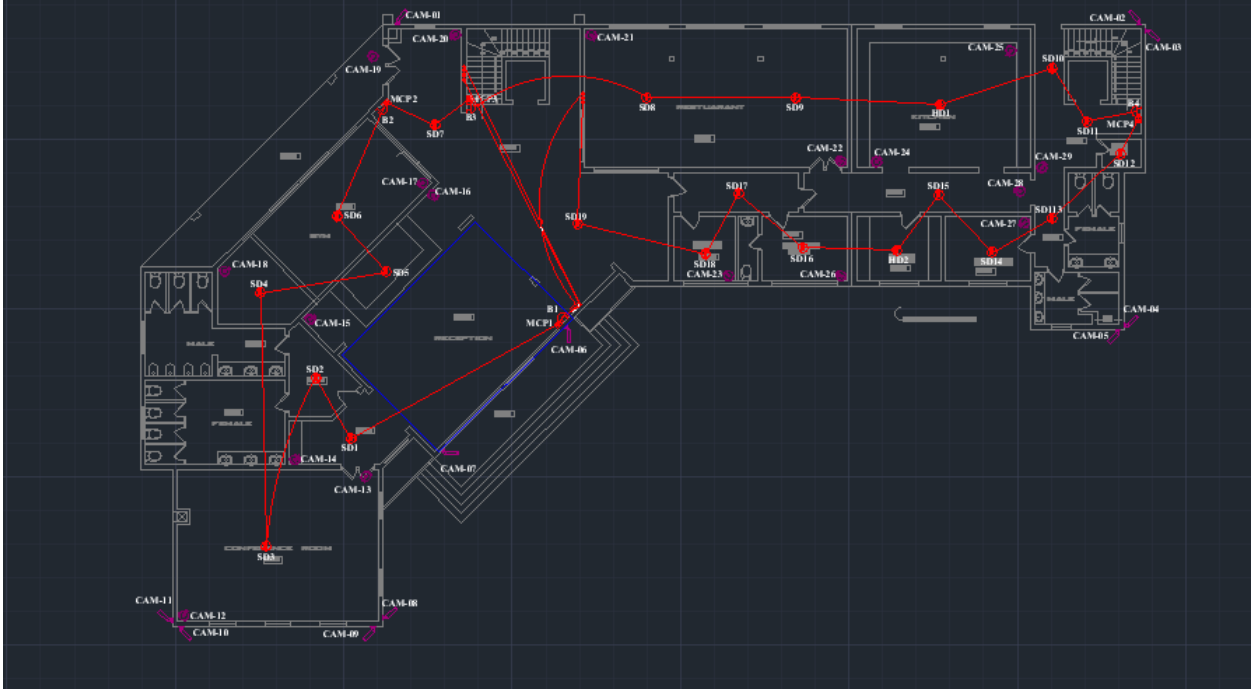
- Electrical Power Layouts
- Lighting Layouts
- Fire Alarm Layouts
- Fire Alarm Schematics
- CCTV Layouts
- Consumer Unit (CU) Arrangements
- Circuit Distribution Drawings
- Electrical Load Schedules

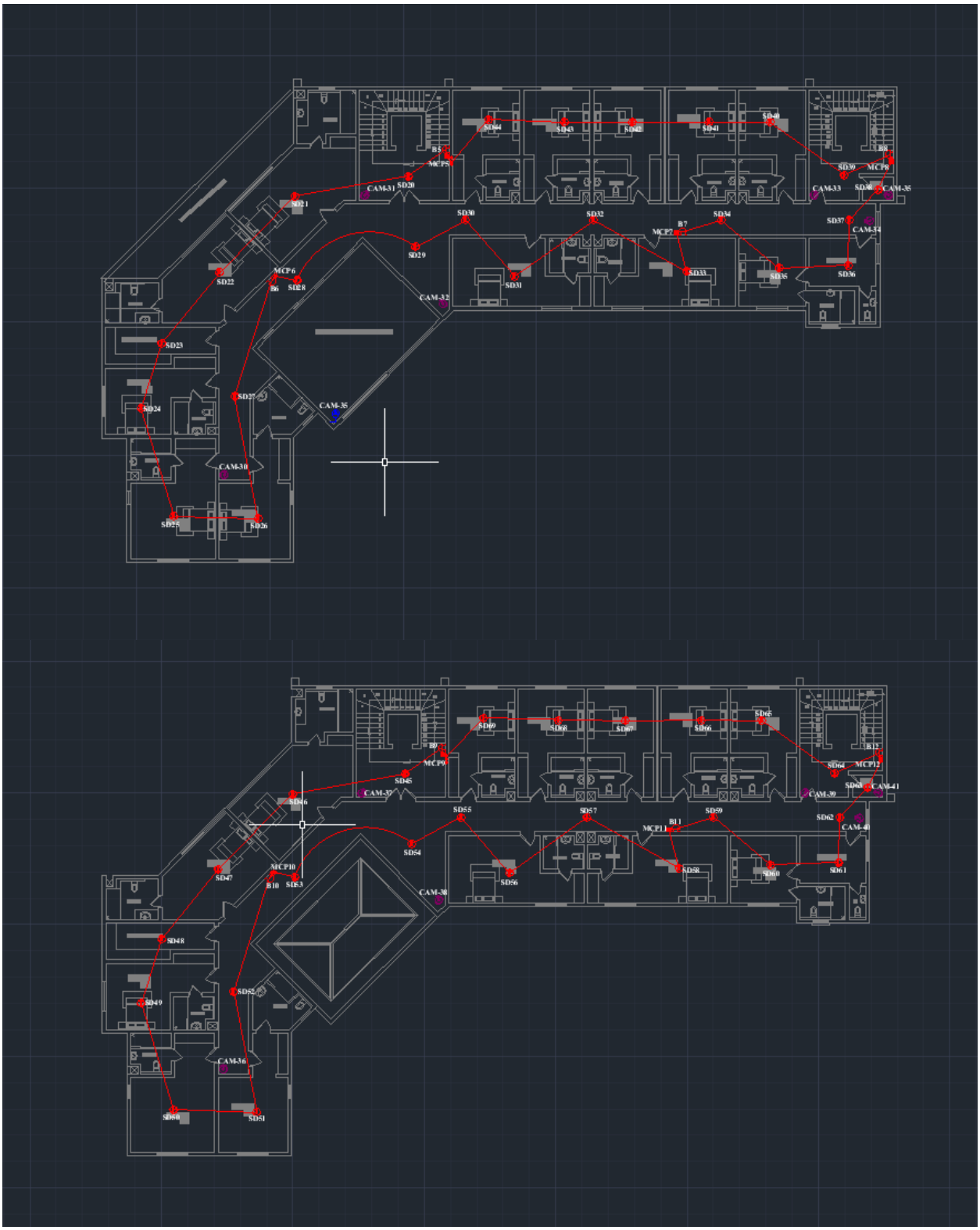
PROJECT 1

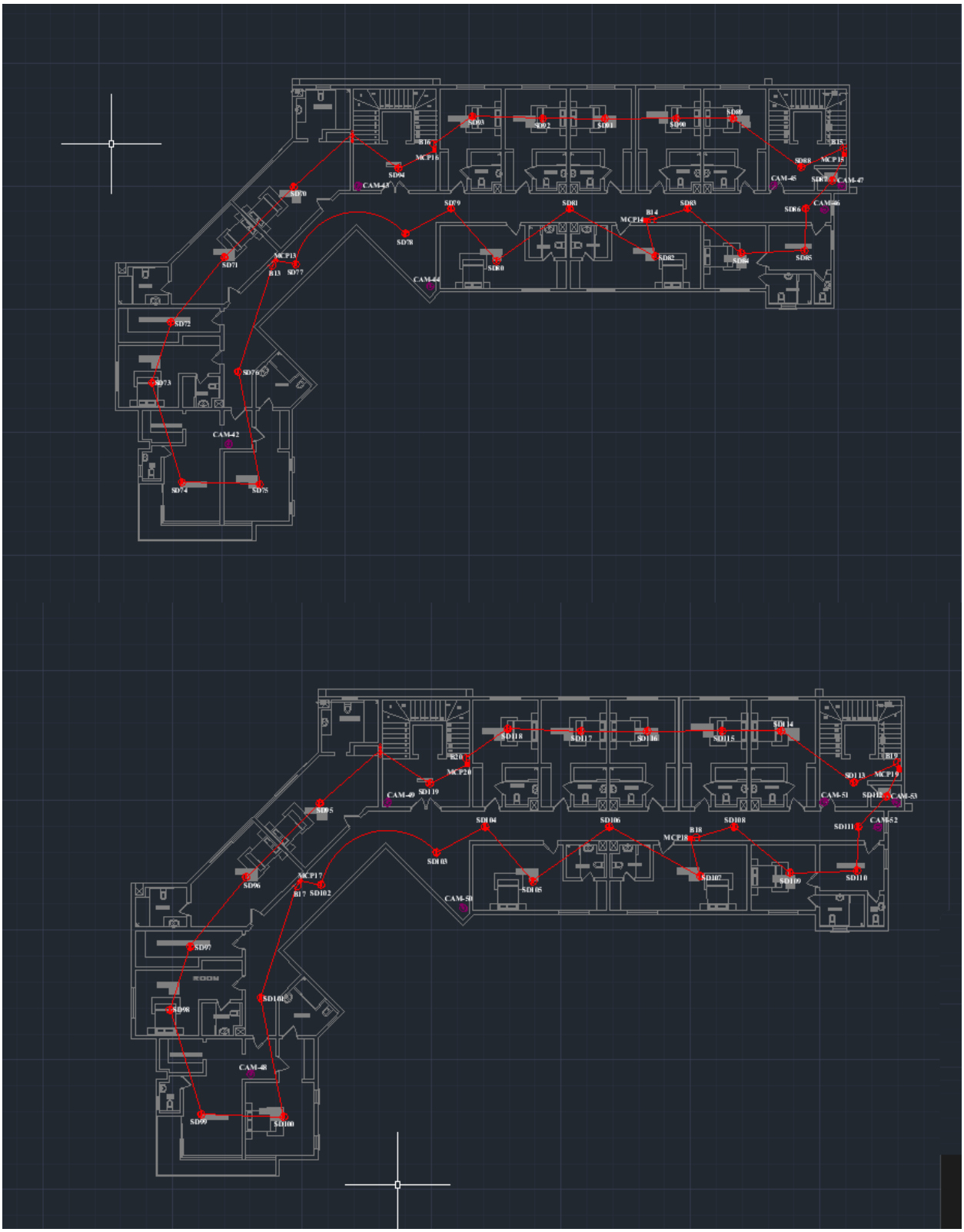
Fire Alarm and CCTV System Design – 5-Storey Ho

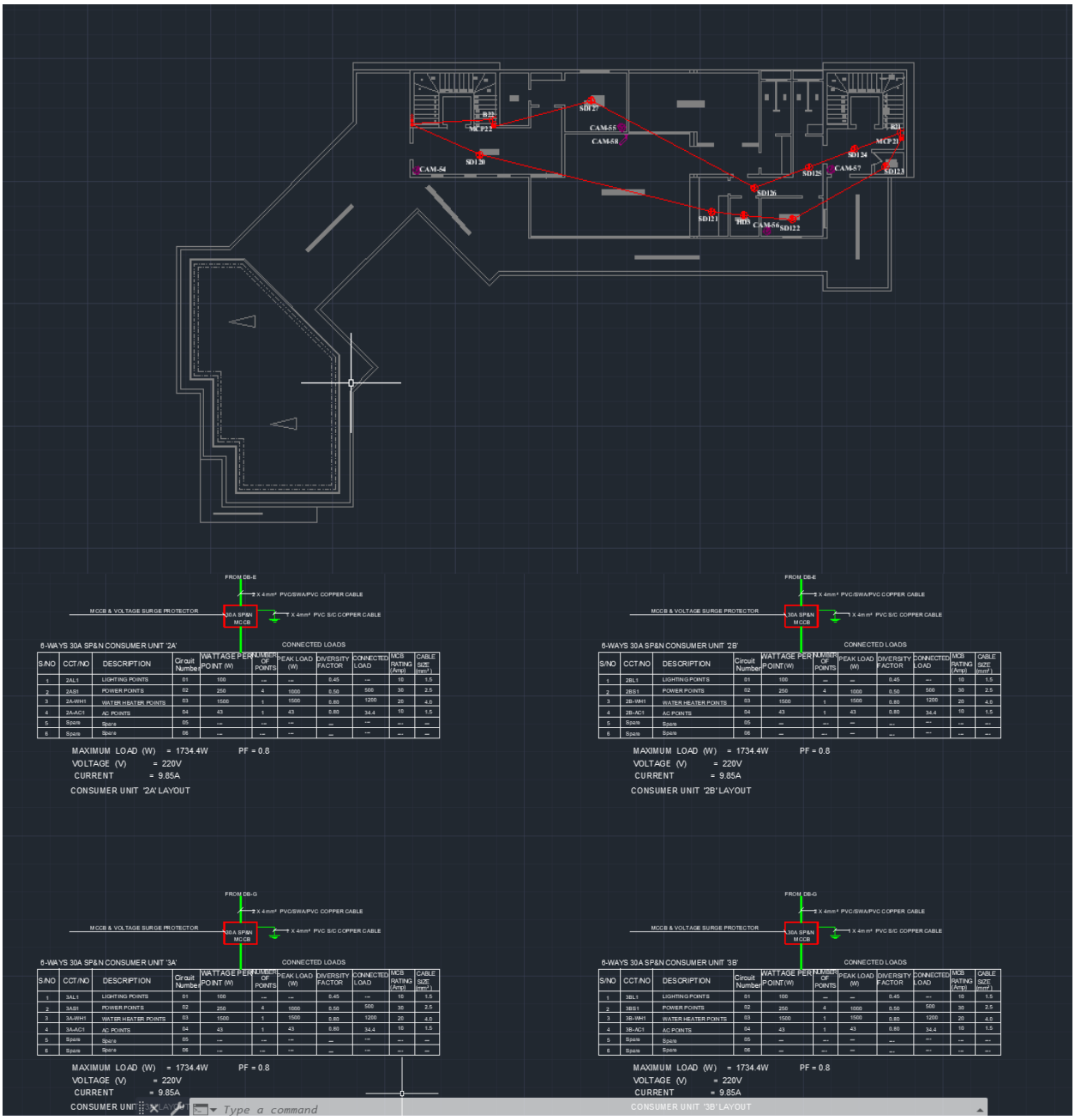



FACP
2 Loop Addressable Fire Alarm Panel
Fire Detection & Alarm Loop
1 x 2 x 1.5mm²
LOOP 1 GROUND FLOOR
Fire Detection & Alarm Loop
1 x 2 x 1.5mm²
LOOP 2 GROUND FLOOR


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RESTAURANT
RECEPTION
CONFERENCE ROOM
MALE
FEMALE


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CAM-52
CAM-53
ROOM



FROM DB-E

X 4mm² PVC/SWA/PVC COPPER CABLE

MCCB & VOLTAGE SURGE PROTECTOR

63A SP&N MCCB

X 4mm² PVC S/C COPPER CABLE

6-WAYS 30A SP&N CONSUMER UNIT '2A'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
						CONNECTED LOADS				
1	2AL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	2AS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	2A-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	2A-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8

VOLTAGE (V) = 220V

CURRENT = 9.85A

CONSUMER UNIT '2A' LAYOUT

FROM DB-E

X 4mm² PVC/SWA/PVC COPPER CABLE

MCCB & VOLTAGE SURGE PROTECTOR

63A SP&N MCCB

X 4mm² PVC S/C COPPER CABLE

6-WAYS 30A SP&N CONSUMER UNIT '2B'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
						CONNECTED LOADS				
1	2BL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	2BS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	2B-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	2B-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8

VOLTAGE (V) = 220V

CURRENT = 9.85A

CONSUMER UNIT '2B' LAYOUT

FROM DB-G

X 4mm² PVC/SWA/PVC COPPER CABLE

MCCB & VOLTAGE SURGE PROTECTOR

63A SP&N MCCB

X 4mm² PVC S/C COPPER CABLE

6-WAYS 30A SP&N CONSUMER UNIT '3A'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
						CONNECTED LOADS				
1	3AL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	3AS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	3A-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	3A-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8

VOLTAGE (V) = 220V

CURRENT = 9.85A

CONSUMER UNIT '3A' LAYOUT

FROM DB-G

X 4mm² PVC/SWA/PVC COPPER CABLE

MCCB & VOLTAGE SURGE PROTECTOR

63A SP&N MCCB

X 4mm² PVC S/C COPPER CABLE

6-WAYS 30A SP&N CONSUMER UNIT '3B'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
						CONNECTED LOADS				
1	3BL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	3BS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	3B-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	3B-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

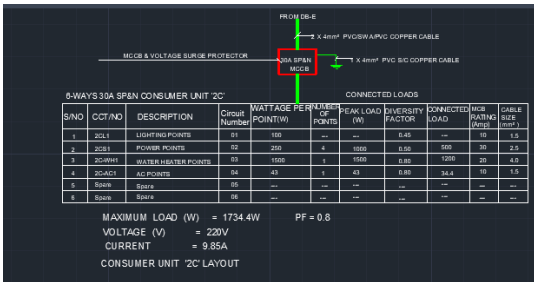
MAXIMUM LOAD (W) = 1734.4W PF = 0.8

VOLTAGE (V) = 220V

CURRENT = 9.85A

CONSUMER UNIT '3B' LAYOUT

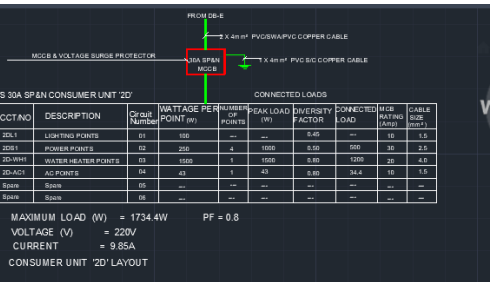
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6-WAYS 30A SP&N CONSUMER UNIT '2C'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT(W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	2CL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	2CS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	2C-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	2C-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

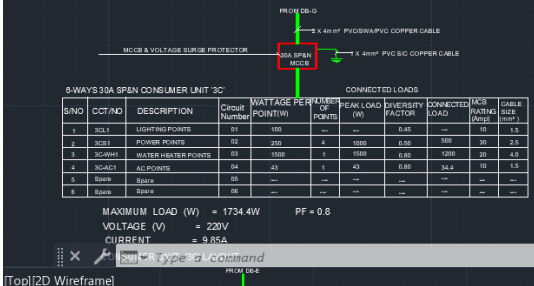
MAXIMUM LOAD (W) = 1734.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 9.85A
CONSUMER UNIT '2C' LAYOUT



6-WAYS 30A SP&N CONSUMER UNIT '2D'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	2DL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	2DS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	2D-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	2D-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

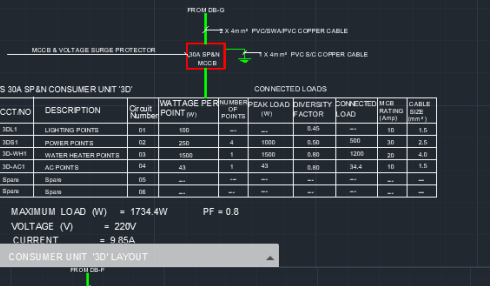
MAXIMUM LOAD (W) = 1734.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 9.85A
CONSUMER UNIT '2D' LAYOUT



6-WAYS 30A SP&N CONSUMER UNIT '3C'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT(W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	3CL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	3CS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	3C-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	3C-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 9.85A



6-WAYS 30A SP&N CONSUMER UNIT '3D'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	3DL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	3DS1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	3D-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	3D-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 9.85A
CONSUMER UNIT '3D' LAYOUT

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[Top][2D Wireframe]



6-WAYS 30A SP&N CONSUMER UNIT '2E'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT(W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	2EL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	2ES1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	2E-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	2E-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 9.85A
CONSUMER UNIT '2E' LAYOUT



6-WAYS 30A SP&N CONSUMER UNIT '2F'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	2FL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	2FS1	POWER POINTS	02	250	8	2000	0.50	1000	30	2.5
3	2F-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	2F-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	2F-AC2	AC POINTS	05	45	1	45	0.80	36	10	1.5
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 2270.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 12.90A
CONSUMER UNIT '2F' LAYOUT



6-WAYS 30A SP&N CONSUMER UNIT '3E'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT(W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	3EL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	3ES1	POWER POINTS	02	250	4	1000	0.50	500	30	2.5
3	3E-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	3E-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1734.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 9.85A
CONSUMER UNIT '3E' LAYOUT

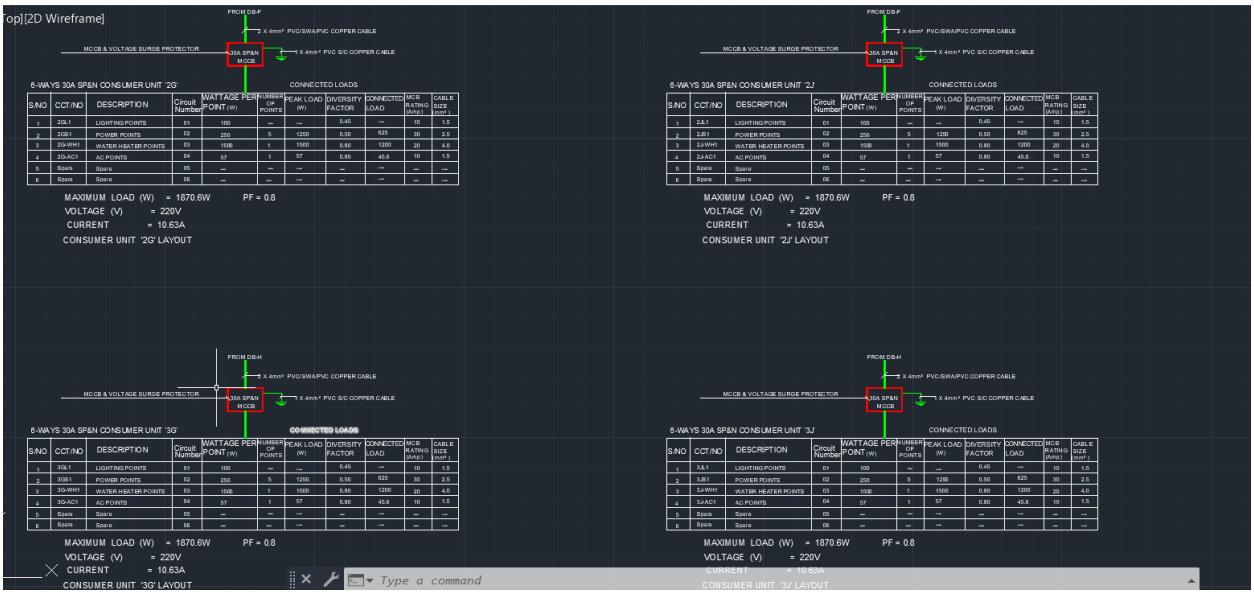


6-WAYS 30A SP&N CONSUMER UNIT '3F'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
					CONNECTED LOADS					
1	3FL1	LIGHTING POINTS	01	100	---	---	0.45	---	10	1.5
2	3FS1	POWER POINTS	02	250	8	2000	0.50	1000	30	2.5
3	3F-WH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	20	4.0
4	3F-AC1	AC POINTS	04	43	1	43	0.80	34.4	10	1.5
5	3F-AC2	AC POINTS	05	45	1	45	0.80	36	10	1.5
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 2270.4W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 12.90A
CONSUMER UNIT '3F' LAYOUT

Type a command



Asset Metadata

Item	Description
Project ID	P001
Building Type	Commercial Hotel
Engineering Discipline	Electrical Engineering
Asset Category	ELV Systems
Software	AutoCAD
File Format	DWG
Drawing Status	Completed Design

Project Overview

This project covers the design and drafting of the fire alarm and CCTV systems for a five-storey hotel. The objective was to provide effective fire detection, alarm notification, and surveillance coverage throughout the facility while maintaining an organized and maintainable system layout.

Scope of Work

- Fire alarm layout design for all floors

- CCTV camera layout design
- Fire alarm schematic development
- Consumer Unit (CU) distribution arrangement
- Fire detection zoning
- Device positioning based on architectural layouts

Engineering Assets Produced

- Fire Alarm Layout Plans
- CCTV Layout Plans
- Fire Alarm and CCTV System Schematics
- Consumer Unit Distribution Diagram

Deliverables

- AutoCAD Drawings (.dwg)

Design Considerations

The fire alarm system was developed with emphasis on detector spacing, manual call point accessibility, zoning strategy, and sounder coverage.

The CCTV system was arranged to provide surveillance of entrances, corridors, staircases, lobby areas, and other strategic locations while maintaining efficient cable routing.

Repository Structure

P001_5Storey_Hotel/

├── CAD_Drawing

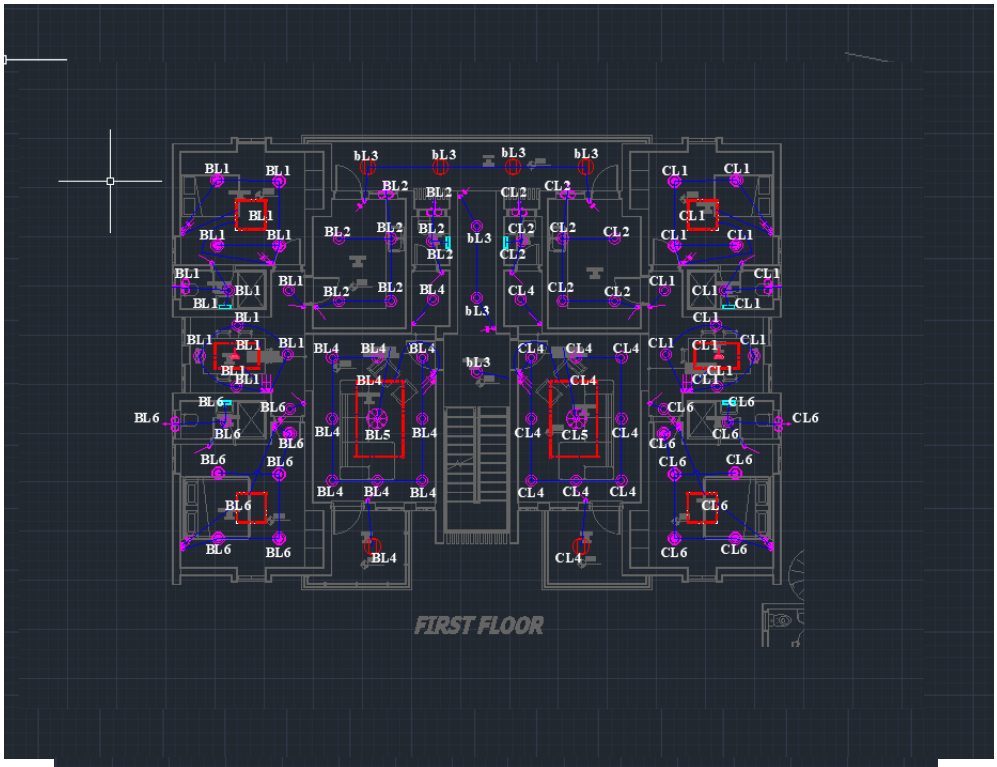
└── README.md

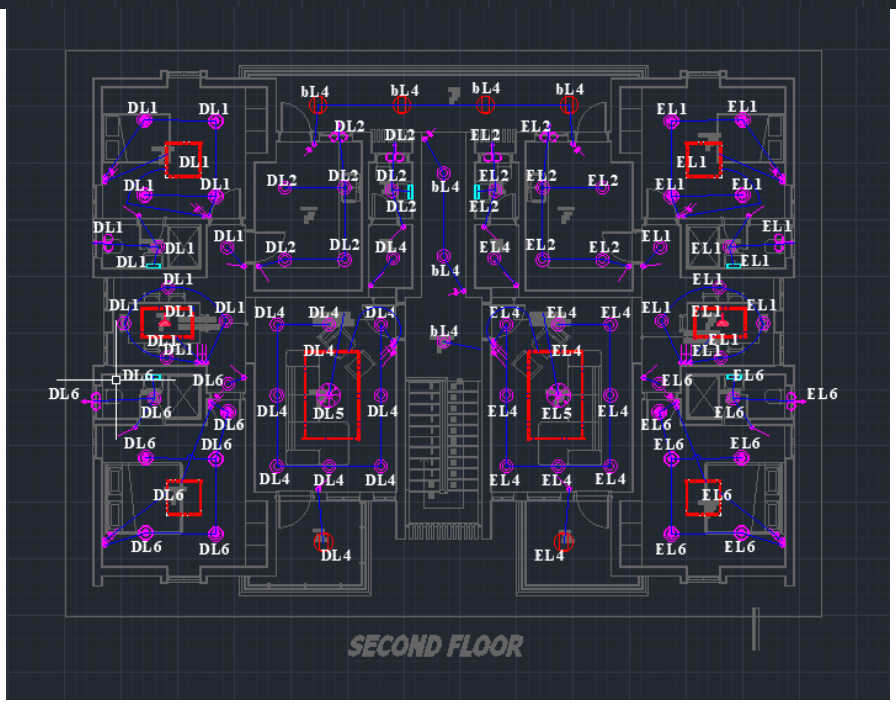
Skills Demonstrated

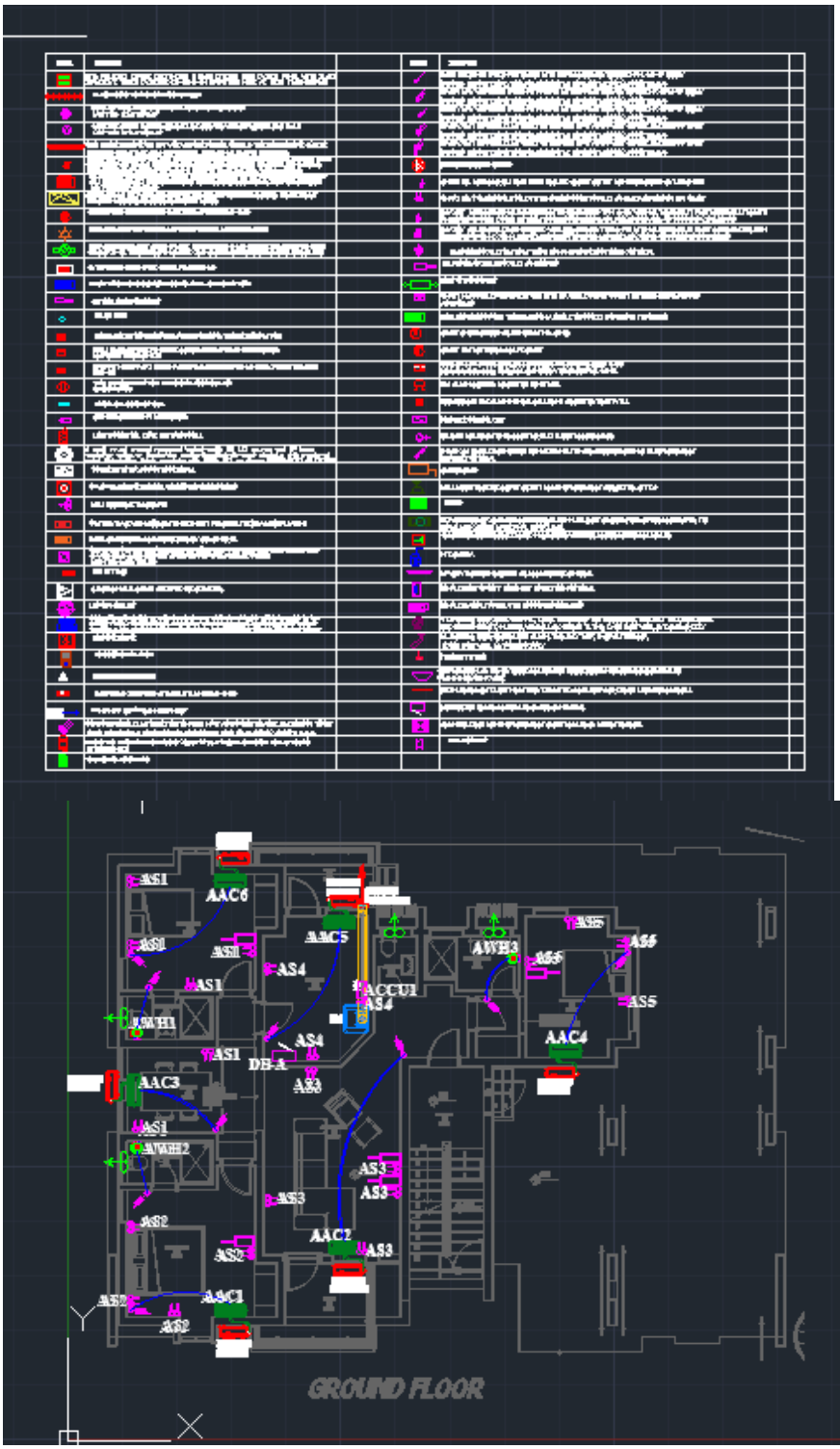
- Commercial electrical drafting
- ELV system planning
- Fire alarm schematic preparation
- CCTV layout planning
- Multi-floor system coordination

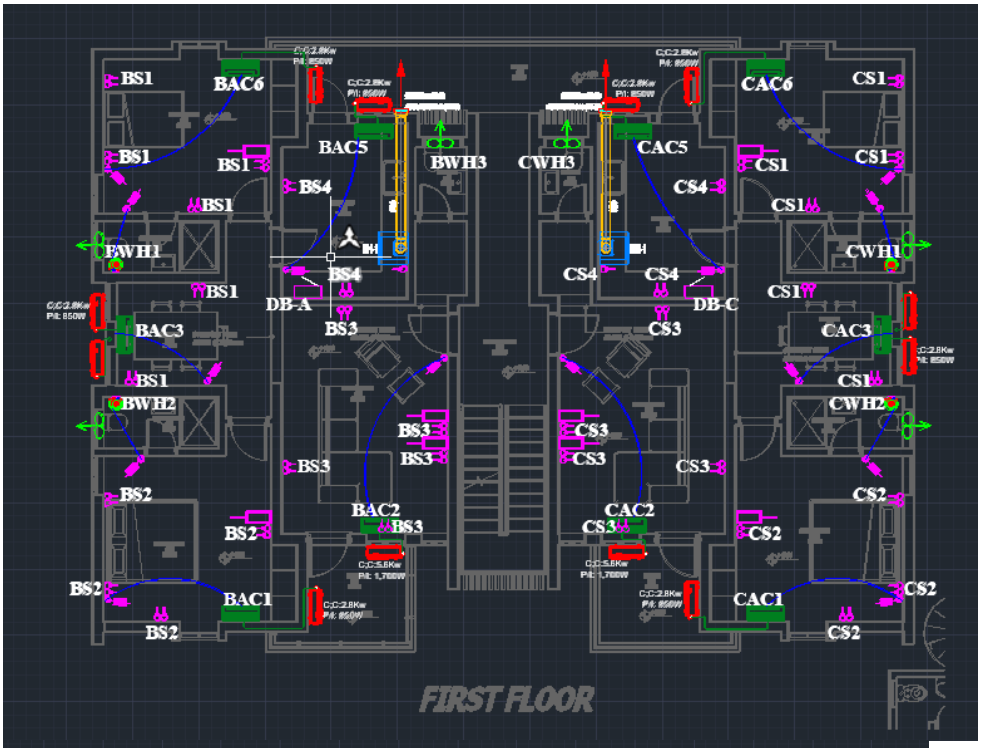
PROJECT 2

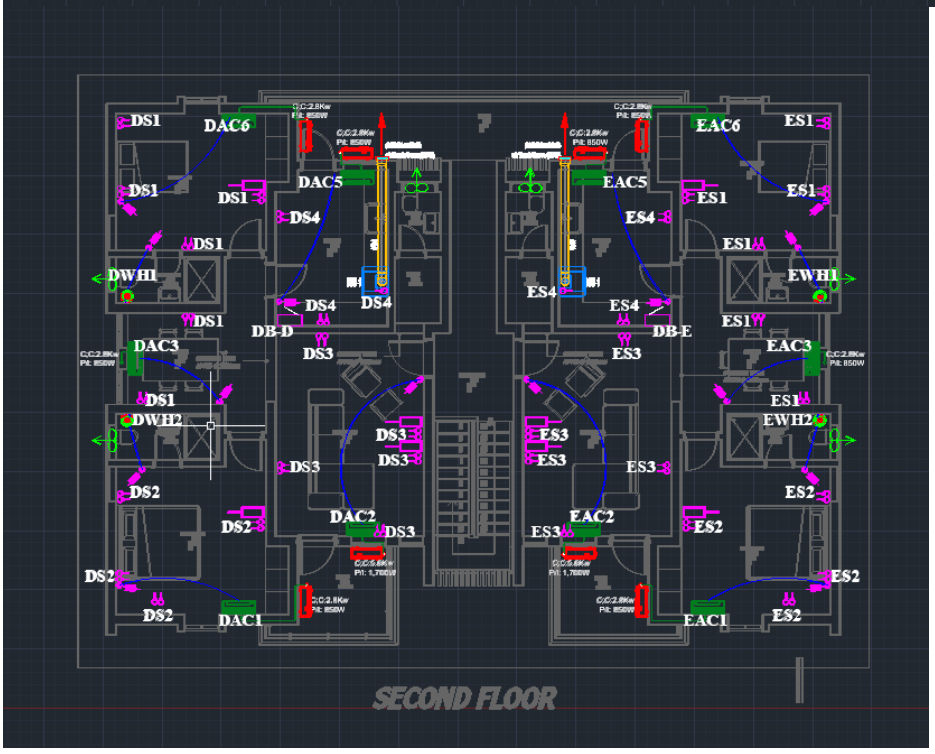
Lighting and Power Design – Residential Apartment






AS1
AAC6
AS1
AS1
AAC5
AS4
AS1
AWH1
AS1
AS4
DB-A
AAC3
AS1
AWH2
AS2
AS2
AS2
AAC1
AS2
AS4
ACCU1
AS4
AWH3
AS5
AS5
AS5
AAC4
AS3
AS3
AS3
AAC2
AS3
GROUND FLOOR


BS1
BAC6
BAC5
BWH3
BS4
EWH1
DB-A
BS3
BAC3
BWH2
BS2
BAC2
BAC1
CAC6
CS1
CAC5
CWH3
CS4
CWH1
DB-C
CS3
CAC3
CWH2
CS2
CAC2
CAC1
FIRST FLOOR


DS1
DAC6
DAC5
DS4
DWH1
DB-D
DS3
DAC3
DWH2
DS2
DAC2
DAC1
EAC6
ES1
EAC5
ES4
EWH1
DB-E
ES3
EAC3
EWH2
ES2
EAC2
EAC1
SECOND FLOOR

GROUND FLOOR DISTRIBUTION BOARD (DB-A)

Location: KITCHEN | **Type:** 63A 8-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELEC
Voltage & Freq.: 230/400V, 50Hz | **DB Ref.:** DB-A
Cable Length(m) to ControlPanel | **Fed from** CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
AS3 - SOCKET POINTS	3 x 2.5	5	250	1250	0.45	562.50			1	1200			0.40	3000	3000	1	3 x 6.0	ACCU1
AL3 - LIGHTING POINTS	2 x 1.5	9	100	900	0.45		405.00		2		495.00		0.45	1100	100	11	2 x 1.5	AL4 - LIGHTING POINTS
AL5 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45			450.00	3			450.00	0.45	1000	100	10	2 x 1.5	AL6 - LIGHTING POINTS
AWH1	3 x 4.0	1	1500	1500	0.80	1200.00			4	1360			0.80	1700	1700	1	3 x 4.0	AAC2 - 20A SWITCH OUTLET
AS1 - SOCKET POINTS	3 x 2.5	6	250	1500	0.45		675.00		5		680		0.80	850	850	1	3 x 4.0	AAC1 - 20A SWITCH OUTLET
AS2 - SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	6			680.00	0.80	850	850	1	3 x 4.0	AAC3 - 20A SWITCH OUTLET
SPARE	3 x 2.5	0	250	0	0.45	0.00			7	680			0.80	850	850	1	3 x 4.0	AAC4 - 20A SWITCH OUTLET
AS4 - SOCKET POINTS	3 x 2.5	3	250	750	0.45		337.5		8		680		0.80	850	850	1	3 x 4.0	AAC5 - 20A SWITCH OUTLET
AL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45			675	9			680	0.80	850	850	1	3 x 4.0	AAC6 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	1500	0	0.80	0			10	0			0.40	0	3000	0	3 x 6.0	SPARE
AWH2	3 x 4.0	1	1500	1500	0.80		1200		11		450		0.45	1000	250	4	3 x 2.5	AS5 - SOCKET POINTS
AWH3	3 x 4.0	1	1500	1500	0.80			1200	12			405	0.45	900	100	9	2 x 1.5	AL2 - LIGHTING POINTS
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	2 x 1.5	0	100	0	0.45			0				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0				0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0				0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				12400		1763	2617.5	2775		3240.00	2305	2215.00		12950				
* RING MAIN FINBL SUB-CIRCUITS																		

63A MCB; 300mA

DB LOAD CABLE SIZE:
4C x 10mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
63A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	**5002.50**	**4922.50**	**4990.00**		
SPARE(W)				**500**	
TOTAL CONNECTED (W)				***15415.00***	W
POWER FACTOR				**0.8**	Cos ϕ
TOTAL CONNECTED (VA)				**19268.75**	VA
CURRENT (A)				**26.9**	A

EARTH WIRE SIZE:
1 x 10mm² PVC SINGLE CORE COPPER CABLE

FIRST FLOOR DISTRIBUTION BOARD (DB-B)

Location: KITCHEN | **Type:** 63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECT
Voltage & Freq.: 230/400V, 50Hz | **DB Ref.:** DB-B
Cable Length(m) to ControlPanel | **Fed from** CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
BL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45	675.00			1	680			0.80	850	850	1	3 x 4.0	BAC4 - 20A SWITCH OUTLET
BWH2	3 x 4.0	1	1500	1500	0.80		1200.00		2		495.00		0.45	1100	100	11	2 x 1.5	BL4 - LIGHTING POINTS
BL5 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45			450.00	3			450.00	0.45	1000	100	10	2 x 1.5	BL6 - LIGHTING POINTS
BWH1	3 x 4.0	1	1500	1500	0.80	1200.00			4	360			0.45	800	100	8	2 x 1.5	BL2 - LIGHTING POINTS
BAC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		5		680		0.80	850	850	1	3 x 4.0	BAC1 - 20A SWITCH OUTLET
BS2 - SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	6			337.50	0.45	750	250	3	3 x 2.5	BS4 - SOCKET POINTS
BS1 - SOCKET POINTS	3 x 2.5	6	250	1500	0.45	675.00			7	0			0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	850	0	0.80		0		8		680		0.80	850	850	1	3 x 4.0	BAC5 - 20A SWITCH OUTLET
BS3 - SOCKET POINTS	3 x 2.5	5	250	1250	0.45			562.5	9			1360	0.80	1700	1700	1	3 x 4.0	BAC2 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.40	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	1500	0	0.80		0				0		0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45			0				0	0.45	0	100	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1700	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0				0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0				0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				10100		2550	1880	1462.5		1040.00	1855	2147.50		7900				
* RING MAIN FINBL SUB-CIRCUITS																		

63A MCB; 300mA

DB LOAD CABLE SIZE:
4C x 10mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
63A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	**3590.00**	**3735.00**	**3610.00**		
SPARE(W)				**500**	
TOTAL CONNECTED (W)				***11435.00***	W
POWER FACTOR				**0.8**	Cos ϕ
TOTAL CONNECTED (VA)				**14293.75**	VA
CURRENT (A)				**19.9**	A

EARTH WIRE SIZE:
1 x 10mm² PVC SINGLE CORE COPPER CABLE

FIRST FLOOR DISTRIBUTION BOARD (DB-C)

Location: KITCHEN | **Type:** 63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.: 230/400V, 50Hz | **DB Ref.:** DB-C
Cable Length(m) to ControlPanel | **Fed from** CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
CL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45	675.00			1	680			0.80	850	850	1	3 x 4.0	CAC4 - 20A SWITCH OUTLET
CWH2	3 x 4.0	1	1500	1500	0.80		1200.00		2		495.00		0.45	1100	100	11	2 x 1.5	CL4 - LIGHTING POINTS
CL5 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45			450.00	3			450.00	0.45	1000	100	10	2 x 1.5	CL6 - LIGHTING POINTS
CWH1	3 x 4.0	1	1500	1500	0.80	1200.00			4	360			0.45	800	100	8	2 x 1.5	CL2 - LIGHTING POINTS
CAC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		5		680		0.80	850	850	1	3 x 4.0	CAC1 - 20A SWITCH OUTLET
CS2- SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	6			337.50	0.45	750	250	3	3 x 2.5	CS4 - SOCKET POINTS
CS1 - SOCKET POINTS	3 x 2.5	6	250	1500	0.45	675.00			7	0			0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	850	0	0.80		0		8		680		0.80	850	850	1	3 x 4.0	CAC5 - 20A SWITCH OUTLET
CS3 - SOCKET POINTS	3 x 2.5	5	250	1250	0.45			562.5	9			1360	0.80	1700	1700	1	3 x 4.0	CAC2 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.40	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	1500	0	0.80		0				0		0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45			0				0	0.45	0	100	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1700	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0				0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0				0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				10100		2550	1880	1462.5		1040.00	1855	2147.50		7900				
* RING MAIN FINBL SUB-CIRCUITS																		

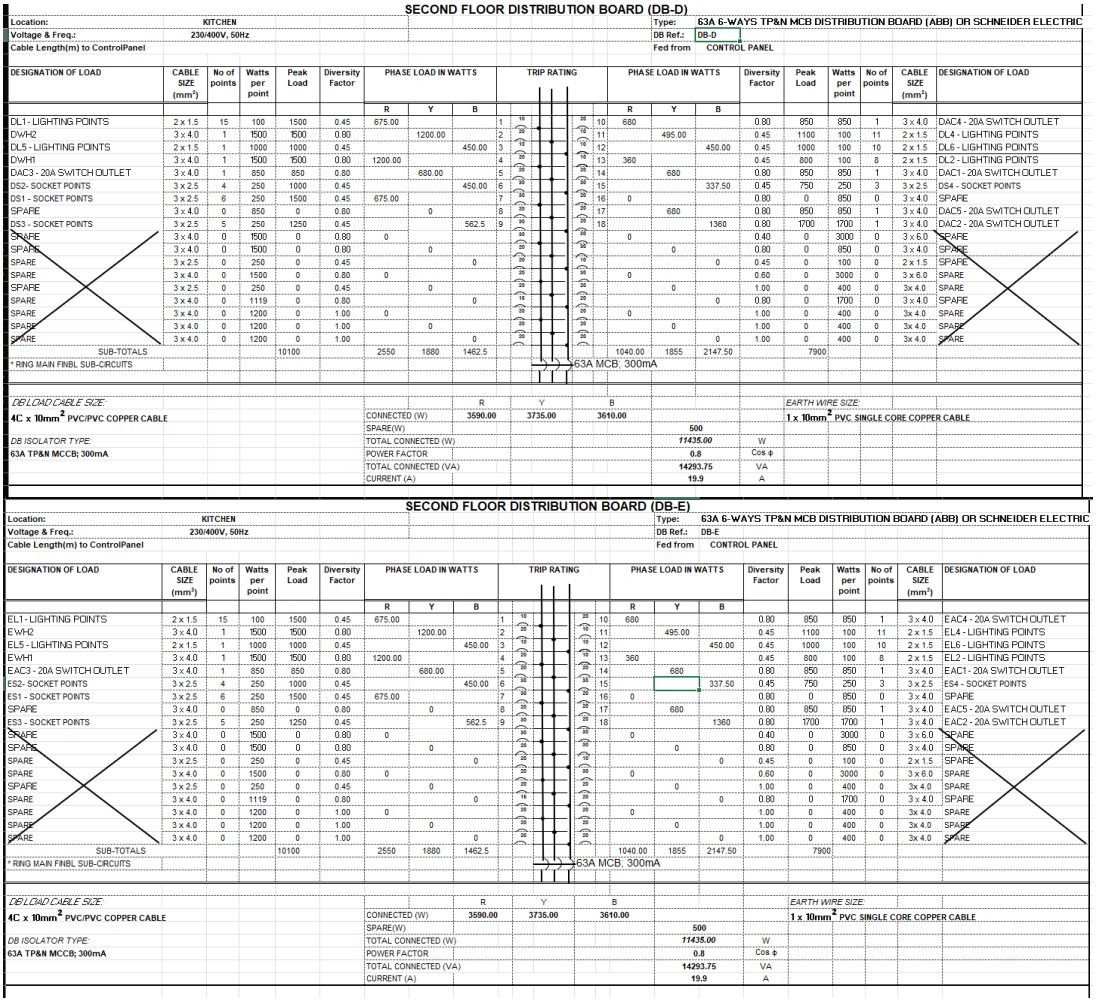
63A MCB; 300mA

DB LOAD CABLE SIZE:
4C x 10mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
63A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	**3590.00**	**3735.00**	**3610.00**		
SPARE(W)				**500**	
TOTAL CONNECTED (W)				***11435.00***	W
POWER FACTOR				**0.8**	Cos ϕ
TOTAL CONNECTED (VA)				**14293.75**	VA
CURRENT (A)				**19.9**	A

EARTH WIRE SIZE:
1 x 10mm² PVC SINGLE CORE COPPER CABLE



SECOND FLOOR DISTRIBUTION BOARD (DB-D)

Location:	KITCHEN	Type:	63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	DB Ref.:	DB-D
Cable Length(m) to ControlPanel		Fed from	CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
DL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45	675.00			1	680			0.80	850	850	1	3 x 4.0	DAC4 - 20A SWITCH OUTLET
DWH2	3 x 4.0	1	1500	1500	0.80		1200.00		2		495.00		0.45	1100	100	11	2 x 1.5	DL4 - LIGHTING POINTS
DL5 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45			450.00	3			450.00	0.45	1000	100	10	2 x 1.5	DL6 - LIGHTING POINTS
DWH1	3 x 4.0	1	1500	1500	0.80	1200.00			4	360			0.45	800	100	8	2 x 1.5	DL2 - LIGHTING POINTS
DAC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		5		680		0.80	850	850	1	3 x 4.0	DAC1 - 20A SWITCH OUTLET
DS2- SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	6			337.50	0.45	750	250	3	3 x 2.5	DS4 - SOCKET POINTS
DS1 - SOCKET POINTS	3 x 2.5	6	250	1500	0.45	675.00			7	0			0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	850	0	0.80		0		8		680		0.80	850	850	1	3 x 4.0	DAC5 - 20A SWITCH OUTLET
DS3 - SOCKET POINTS	3 x 2.5	5	250	1250	0.45			562.5	9			1360	0.80	1700	1700	1	3 x 4.0	DAC2 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.40	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	1500	0	0.80		0				0		0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45			0				0	0.45	0	100	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1700	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0				0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0				0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				10100		2550	1880	1462.5		1040.00	1855	2147.50		7900				
* RING MAIN FINBL SUB-CIRCUITS																		

63A MCB, 300mA

DB LOAD CABLE SIZE:		R	Y	B	EARTH WIRE SIZE:
4C x 10mm² PVC/PVC COPPER CABLE	CONNECTED (W)	3590.00	3735.00	3610.00	1 x 10mm² PVC SINGLE CORE COPPER CABLE
	SPARE(W)			500	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)			11435.00	W
63A TP&N MCCB; 300mA	POWER FACTOR			0.8	Cos ϕ
	TOTAL CONNECTED (VA)			14293.75	VA
	CURRENT (A)			19.9	A

SECOND FLOOR DISTRIBUTION BOARD (DB-E)

Location:	KITCHEN	Type:	63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	DB Ref.:	DB-E
Cable Length(m) to ControlPanel		Fed from	CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
EL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45	675.00			1	680			0.80	850	850	1	3 x 4.0	EAC4 - 20A SWITCH OUTLET
EWH2	3 x 4.0	1	1500	1500	0.80		1200.00		2		495.00		0.45	1100	100	11	2 x 1.5	EL4 - LIGHTING POINTS
EL5 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45			450.00	3			450.00	0.45	1000	100	10	2 x 1.5	EL6 - LIGHTING POINTS
EWH1	3 x 4.0	1	1500	1500	0.80	1200.00			4	360			0.45	800	100	8	2 x 1.5	EL2 - LIGHTING POINTS
EAC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		5		680		0.80	850	850	1	3 x 4.0	EAC1 - 20A SWITCH OUTLET
ES2- SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	6			337.50	0.45	750	250	3	3 x 2.5	ES4 - SOCKET POINTS
ES1 - SOCKET POINTS	3 x 2.5	6	250	1500	0.45	675.00			7	0			0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	850	0	0.80		0		8		680		0.80	850	850	1	3 x 4.0	EAC5 - 20A SWITCH OUTLET
ES3 - SOCKET POINTS	3 x 2.5	5	250	1250	0.45			562.5	9			1360	0.80	1700	1700	1	3 x 4.0	EAC2 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.40	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	1500	0	0.80		0				0		0.80	0	850	0	3 x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45			0				0	0.45	0	100	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0				0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 2.5	0	250	0	0.45		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1700	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0				0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0				0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0				0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				10100		2550	1880	1462.5		1040.00	1855	2147.50		7900				
* RING MAIN FINBL SUB-CIRCUITS																		

63A MCB, 300mA

DB LOAD CABLE SIZE:		R	Y	B	EARTH WIRE SIZE:
4C x 10mm² PVC/PVC COPPER CABLE	CONNECTED (W)	3590.00	3735.00	3610.00	1 x 10mm² PVC SINGLE CORE COPPER CABLE
	SPARE(W)			500	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)			11435.00	W
63A TP&N MCCB; 300mA	POWER FACTOR			0.8	Cos ϕ
	TOTAL CONNECTED (VA)			14293.75	VA
	CURRENT (A)			19.9	A

Asset Metadata

Item	Description
Project ID	P002
Building Type	Residential Apartment
Engineering Discipline	Electrical Engineering
Asset Category	Lighting & Power
Software	AutoCAD, Microsoft Excel
File Formats	DWG, XLSX

Project Overview

This project presents the complete lighting and power distribution design for a residential apartment. The work includes electrical layouts, load calculations, circuit planning, and consumer unit configuration to ensure safe and balanced electrical distribution.

Scope of Work

- Lighting layout design
- Power layout design
- Load calculations
- Circuit distribution planning
- Load schedule preparation

Engineering Assets Produced

- Lighting Layout Plan
- Power Layout Plan
- Electrical Load Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout	Lighting layout for residential apartment	DWG
P-001	Power Layout	Power layout for residential apartment	DWG
LS-001	Load Schedule	DB for residential apartment	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Excel Load Schedule (.xlsx)

Design Considerations

The lighting system was arranged to achieve balanced illumination and logical circuit grouping. Power distribution was designed to maintain safe circuit loading, proper socket distribution, and effective consumer unit organization. Electrical calculations were used to determine connected load, demand load, protective device ratings, and cable sizing requirements.

Repository Structure

P002_Residential_Apartment/

├── Load_Schedules

├── CAD_Drawings

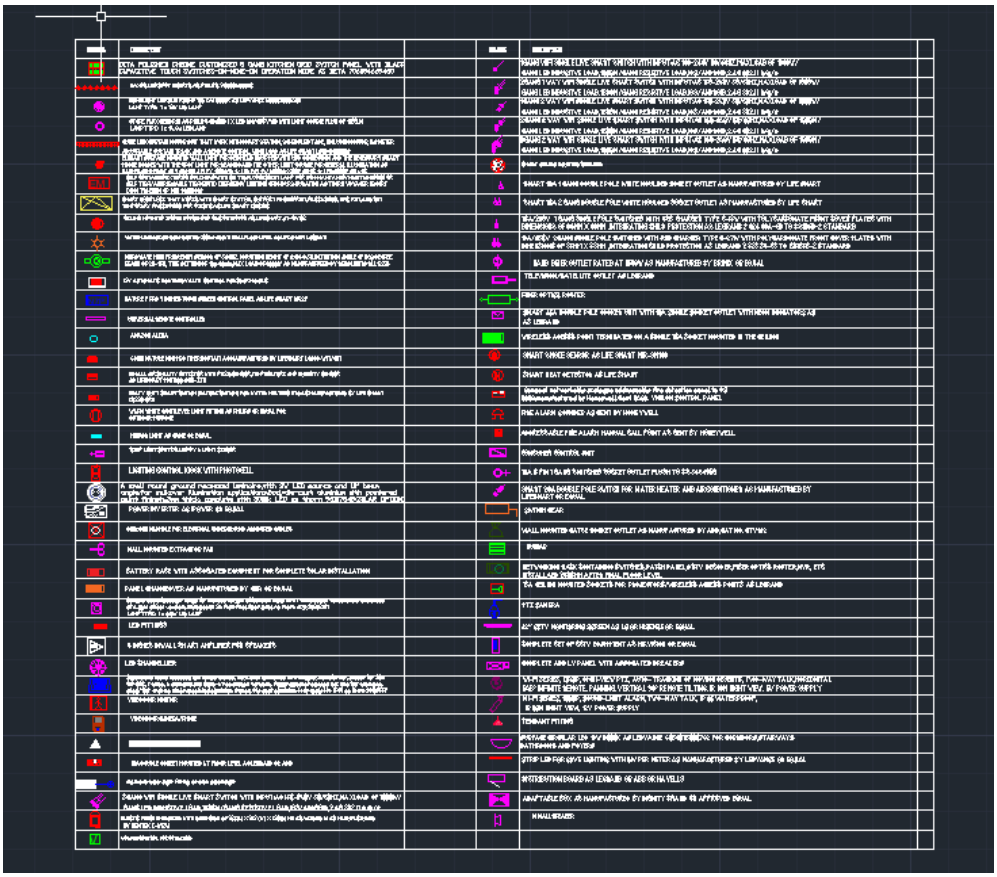
└── README.md

Skills Demonstrated

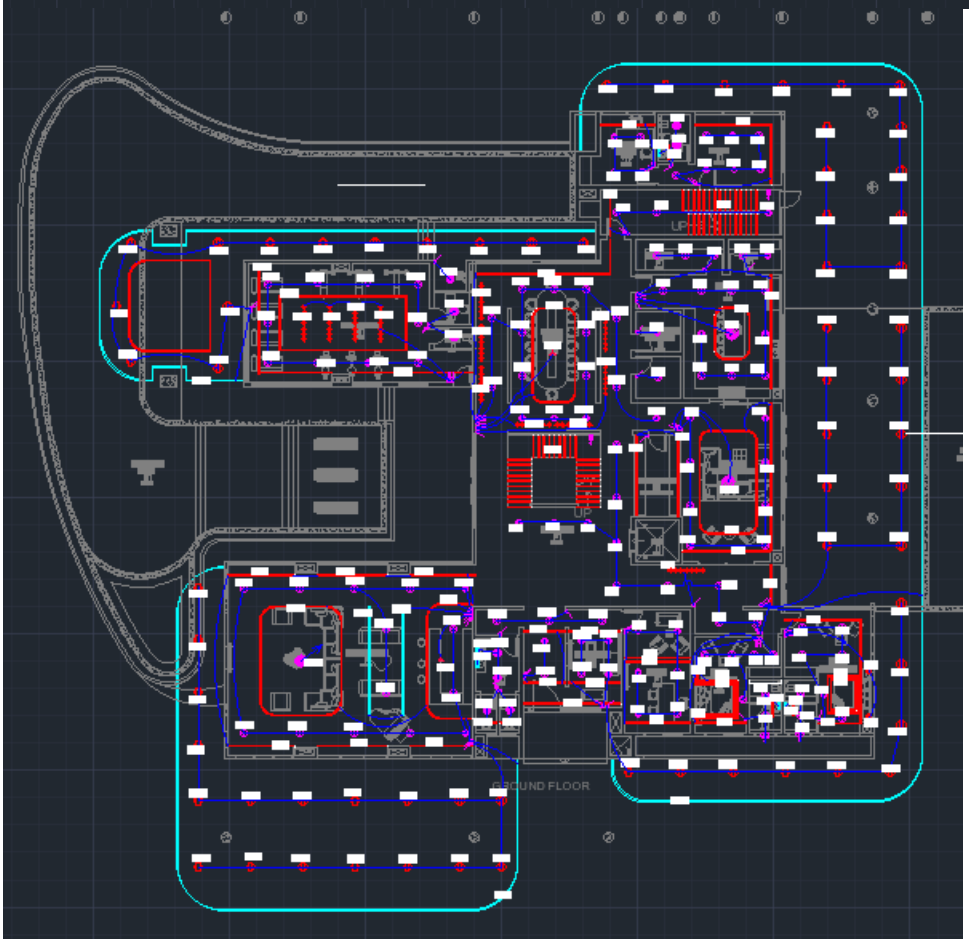
- Residential electrical design
- Lighting layout planning
- Load calculation
- Circuit planning
- Consumer unit configuration

PROJECT 3

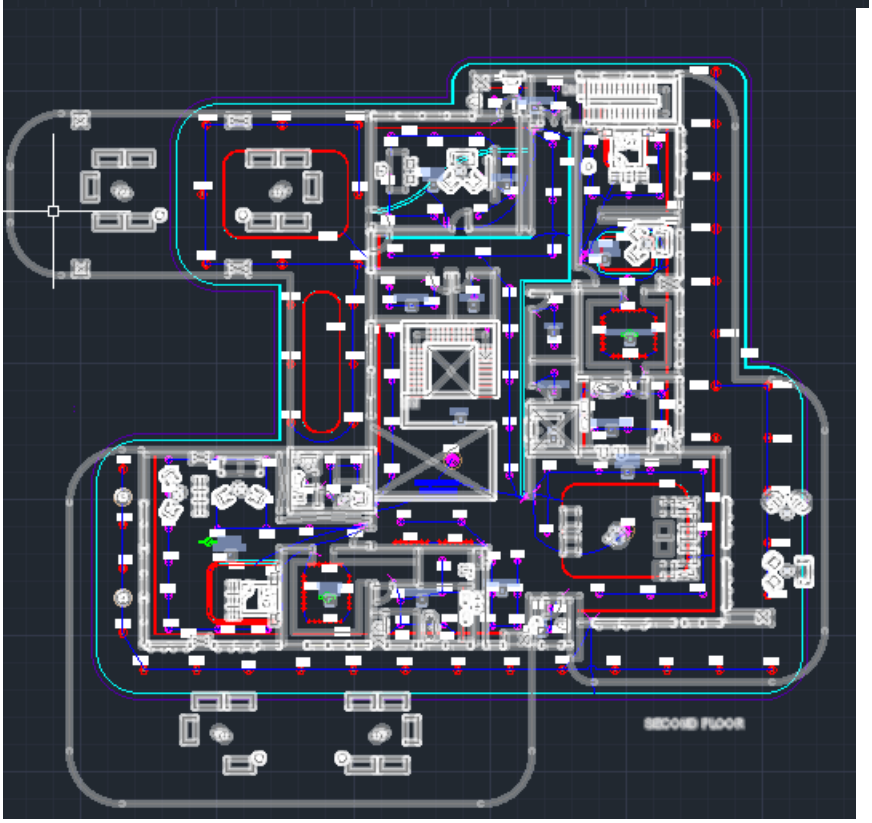
Lighting, Power and ELV Design – 2-Storey Residential Complex

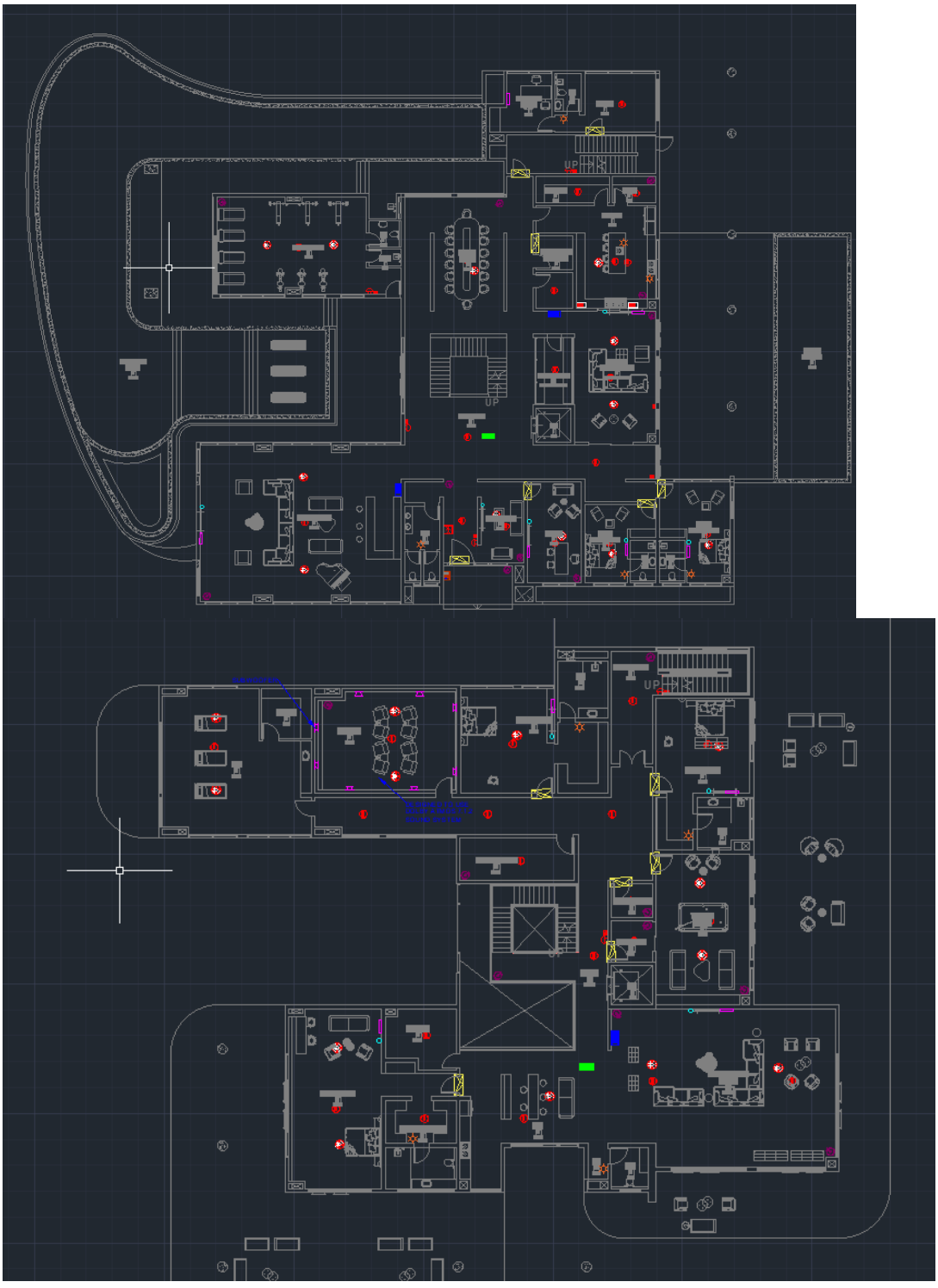


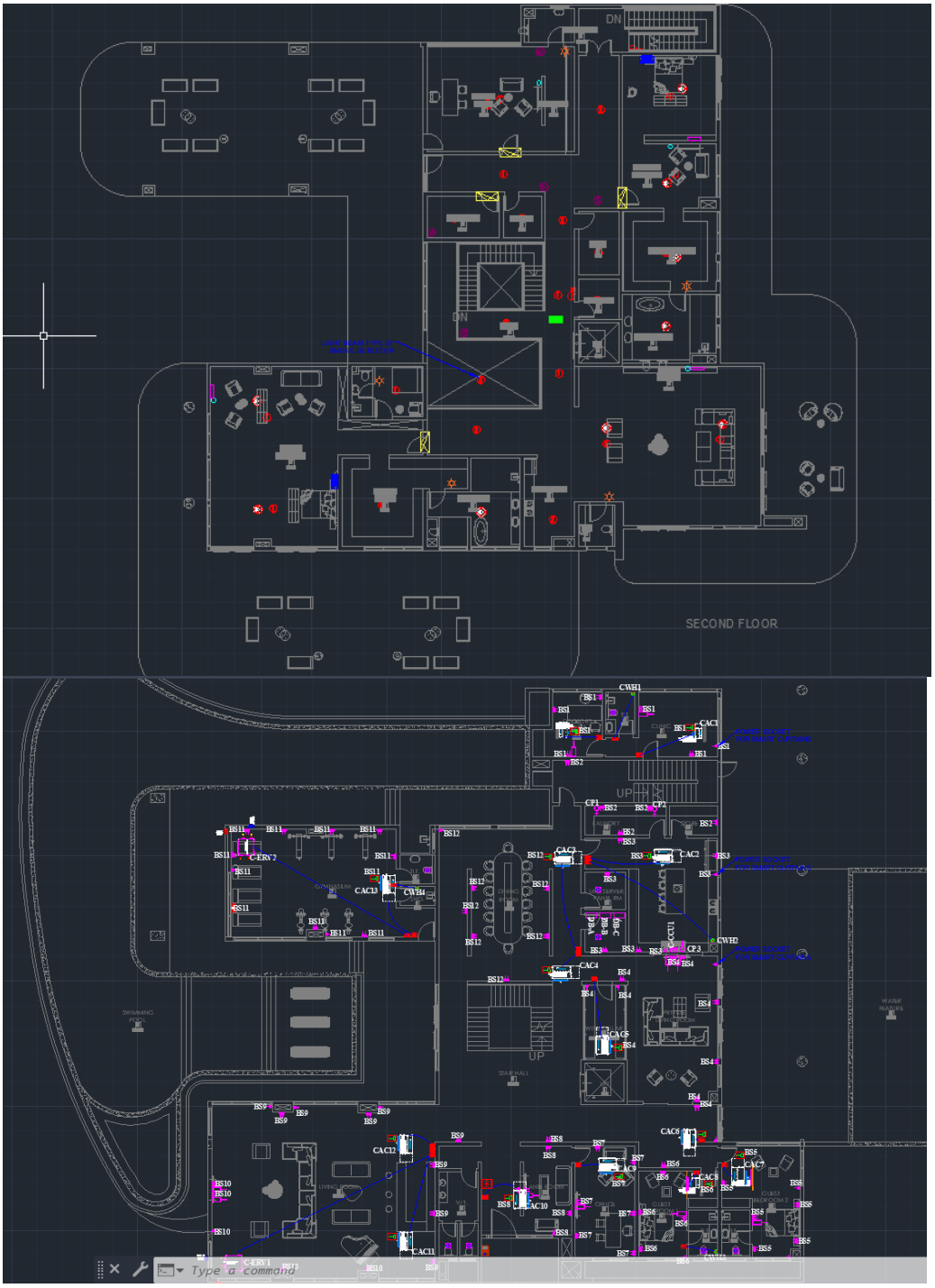
	A	B	C	D
1	LIGHTING			
2	S/N	SYMBOL	DESCRIPTION	QUANTITY
3	1			24
4	2			9
5	3			335
6	4			14
7	5			150
8	6			14
9	7			8
10	8			
11	9		PENDANT FITTING	2
12	10			14
13	11			51
14	12			27
15	13			1

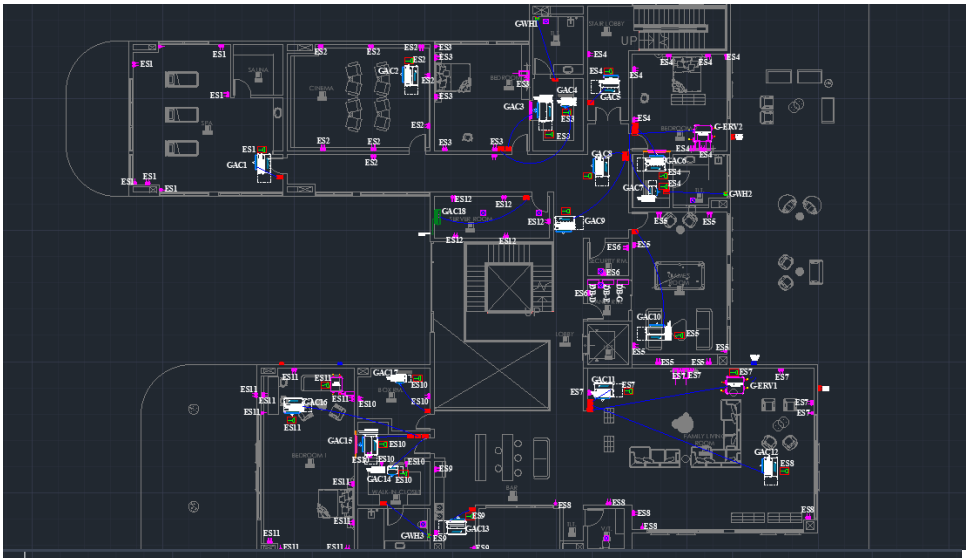


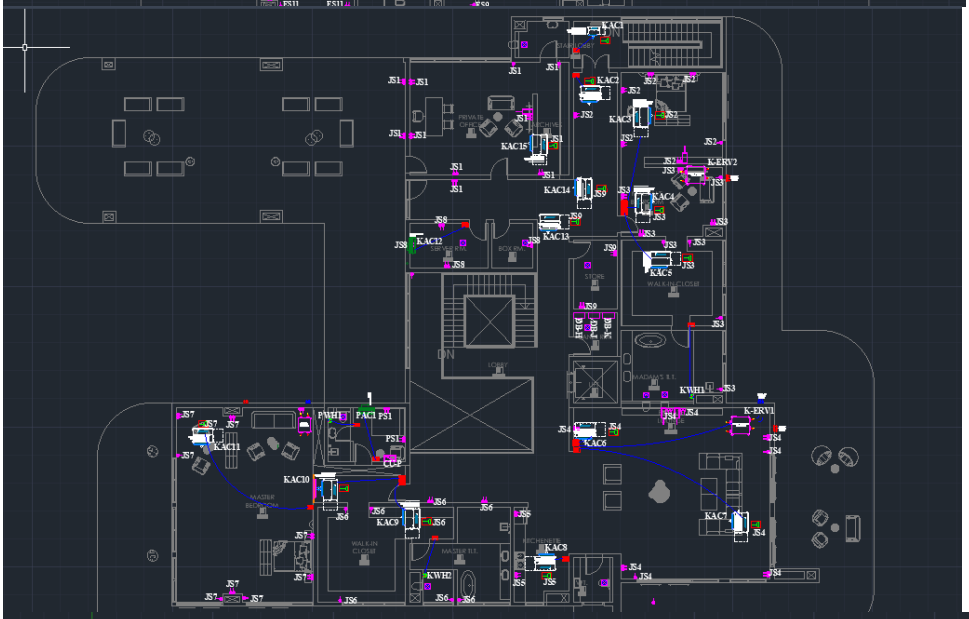

FIRST FLOOR


SECOND FLOOR

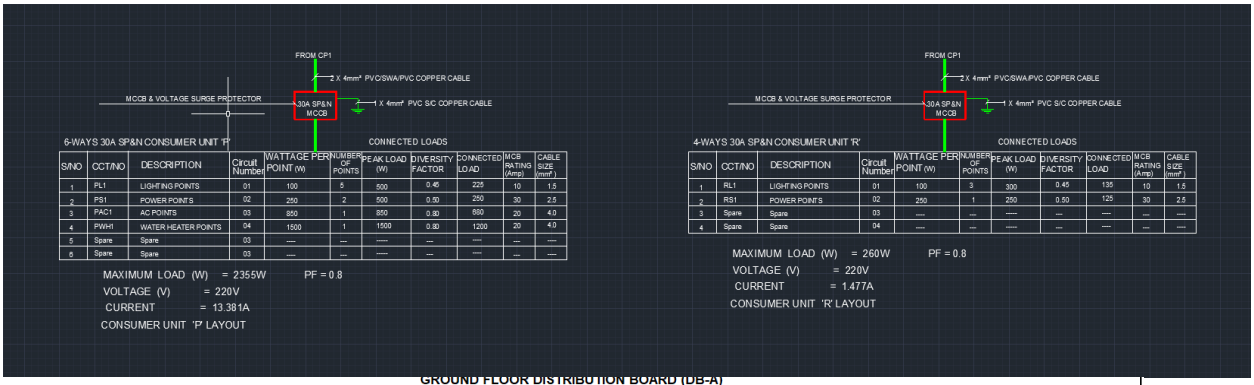



SECOND FLOOR
Type a command





POWER			
S/N	SYMBOL	DESCRIPTION	QUANTITY
1			16
2			1
3			20
4			59
5			152
6			1
7			3
9			47



S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
1	PL1	LIGHTING POINTS	01	100	5	500	0.45	225	10	1.5
2	PS1	POWER POINTS	02	250	2	500	0.50	250	30	2.5
3	PAC1	AC POINTS	03	850	1	850	0.80	680	20	4.0
4	PWH1	WATER HEATER POINTS	04	1500	1	1500	0.80	1200	20	4.0
5	Spare	Spare	03	----	---	-----	---	-----	---	----
6	Spare	Spare	03	----	---	-----	---	-----	---	----

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
1	RL1	LIGHTING POINTS	01	100	3	300	0.45	135	10	1.5
2	RS1	POWER POINTS	02	250	1	250	0.50	125	30	2.5
3	Spare	Spare	03	----	---	-----	---	-----	---	----
4	Spare	Spare	04	----	---	-----	---	-----	---	----

GROUND FLOOR DISTRIBUTION BOARD (DB-A)

Location: STORE
Voltage & Freq.: 230/400V, 50Hz
Cable Length(m) to ControlPanel
Type: 63A 8-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
DB Ref.: DB-A
Fed from CP1

DESIGNATION OF LOAD	CABLE SIZE (mm²)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm²)	DESIGNATION OF LOAD
SPARE	2 x 1.5	0	100	0	0.45	0.00			1	810.00			0.45	1800	100	18	2 x 2.5	AL2 - LIGHTING
AL3 - LIGHTING	2 x 1.5	1	1000	1000	0.45		450.00		2		450.00		0.45	1000	1000	1	2 x 1.5	AL4 - LIGHTING
AL5 - LIGHTING	2 x 2.5	12	100	1200	0.45			540.00	3			585.00	0.45	1300	100	13	2 x 2.5	AL6 - LIGHTING
AL7 - LIGHTING	2 x 2.5	17	100	1700	0.45	765.00			4	585			0.45	1300	100	13	2 x 2.5	AL8 - LIGHTING
AL9 - LIGHTING	2 x 2.5	9	100	900	0.45		405		5		990		0.45	2200	100	22	2 x 2.5	AL10 - LIGHTING
AL11 - LIGHTING	2 x 2.5	23	100	2300	0.45			1035.00	6			855	0.45	1900	100	19	2 x 2.5	AL12 - LIGHTING
AL13 - LIGHTING	2 x 1.5	1	1000	1000	0.45	450.00			7	630			0.45	1400	100	14	2 x 2.5	AL14 - LIGHTING
AL15 - LIGHTING	2 x 2.5	20	100	2000	0.45		900		8		1125		0.45	2500	100	25	2 x 2.5	AL16 - LIGHTING
AL17 - LIGHTING	2 x 1.5	1	1000	1000	0.45			450.00	9			450	0.45	1000	1000	1	2 x 1.5	AL18 - LIGHTING
AL19 - LIGHTING	2 x 2.5	19	100	1900	0.45	855			10	675			0.45	1500	100	15	2 x 2.5	AL20 - LIGHTING
SPARE	2 x 1.5	0	100	0	0.45		0		11	23		0	0.4	0	6000	0	3 x 4.0	SPARE
AL1 - LIGHTING	2 x 2.5	20	100	2000	0.45			900	12	24		0	0.5	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				15000		2070	1755	2925		2700.00	2565	1890.00		15900				
* RING MAIN FINAL SUB-CIRCUITS																		

63A MCB; 300mA

DB LOAD CABLE SIZE:
4C x 6mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
63A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	**4770.00**	**4320**	**4815.00**		
SPARE(W)				**500**	
TOTAL CONNECTED (W)				***14405.00***	W
POWER FACTOR				**0.8**	Cos ϕ
TOTAL CONNECTED (VA)				**18006.25**	VA
CURRENT (A)				**25.1**	A

EARTH WIRE SIZE:
1 x 4mm² PVC SINGLE CORE COPPER CABLE

GROUND FLOOR DISTRIBUTION BOARD (DB-B)

Location: STORE
Voltage & Freq.: 230/400V, 50Hz
Cable Length(m) to ControlPanel
Type: 63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
DB Ref.: DB-B
Fed from CP1

DESIGNATION OF LOAD	CABLE SIZE (mm²)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm²)	DESIGNATION OF LOAD
BS1 - 13A SOCKET OUTLET	3 x 2.5	8	250	2000	0.50	1000.00			1	875.00			0.50	1750	250	7	3 x 2.5	BS6 - 13A SOCKET OUTLET
BS3 - 13A SOCKET OUTLET	3 x 2.5	8	250	2000	0.50		1000.00		2		1250.00		0.50	2500	250	10	3 x 2.5	BS4 - 13A SOCKET OUTLET
BS5 - 13A SOCKET OUTLET	3 x 2.5	7	250	1750	0.50			875.00	3			625.00	0.50	1250	250	5	3 x 2.5	BS2 - 13A SOCKET OUTLET
BS7 - 13A SOCKET OUTLET	3 x 2.5	7	250	1750	0.50	875.00			4	625			0.50	1250	250	5	3 x 2.5	BS8 - 13A SOCKET OUTLET
BS9 - 13A SOCKET OUTLET	3 x 2.5	9	250	2250	0.50		1125		5		625		0.50	1250	250	5	3 x 2.5	BS10 - 13A SOCKET OUTLET
BS11 - 13A SOCKET OUTLET	3 x 2.5	12	250	3000	0.50			1500.00	6			1000	0.50	2000	250	8	3 x 2.5	BS12 - 13A SOCKET OUTLET
SPARE	3 x 2.5	0	250	0	0.50	0.00			7	0			0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50		0		8		0		0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50			0.00	9			0	0.50	0	250	0	3 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45	0				0			0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 1.5	0	100	0	0.45		0				0		0.4	0	6000	0	3 x 4.0	SPARE
SPARE	2 x 2.5	0	100	0	0.45			0				0	0.5	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				12750		1875	2125	2375		1500.00	1875	1625.00		10000				
* RING MAIN FINAL SUB-CIRCUITS																		

63A MCB; 300mA

DB LOAD CABLE SIZE:
4C x 6mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
63A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	**3375.00**	**4000**	**4000.00**		
SPARE(W)				**500**	
TOTAL CONNECTED (W)				***11875.00***	W
POWER FACTOR				**0.8**	Cos ϕ
TOTAL CONNECTED (VA)				**14843.75**	VA
CURRENT (A)				**20.7**	A

EARTH WIRE SIZE:
1 x 4mm² PVC SINGLE CORE COPPER CABLE

GROUND FLOOR DISTRIBUTION BOARD (DB-C)

Location:	STORE	**Type:**	63A 8-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB-C
Cable Length(m) to ControlPanel		**Fed from**	CP1

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
CAC1 - 20A SWITCH OUTLET	3 x 2.5	1	65	65	0.80	52.00			1			448.00	0.80	560	560	1	3 x 2.5	CAC2 - 20A SWITCH OUTLET
CAC3 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80		448.00		2		448.00		0.80	560	560	1	3 x 2.5	CAC4 - 20A SWITCH OUTLET
CAC5 - 20A SWITCH OUTLET	3 x 2.5	1	490	490	0.80			392.00	3			448.00	0.80	560	560	1	3 x 2.5	CAC6 - 20A SWITCH OUTLET
CAC12 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80	448.00			4	52			0.80	65	65	1	3 x 2.5	CAC8 - 20A SWITCH OUTLET
CAC9 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80		92.80		5		92.80		0.80	116	116	1	3 x 2.5	CAC10 - 20A SWITCH OUTLET
CP1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40			800.00	6			92.8	0.80	116	116	1	3 x 2.5	CAC7 - 20A SWITCH OUTLET
CAC13 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80	448.00			7	320			0.80	400	400	1	3 x 2.5	C-ERV1 - ENERGY RECOVERY UNIT
CAC11 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80		448		8		800		0.40	2000	2000	1	3 x 6.0	C-CCU1 - COOKER UNIT
CP2 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40			800.00	9			320	0.80	400	400	1	3 x 4.0	C-ERV2 - ENERGY RECOVERY UNIT
CP3 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40	800			10	0			0.80	0	1500	0	3 x 4.0	SPARE
SPARE	3 x 6.0	0	3000	0	0.40		0		11		0		0.80	0	400	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	400	0	0.80			0	12			0	0.80	0	116	0	3 x 2.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				8911		1748	988.8	1992		820.00	1341	860.80		4777				
* RING MAIN FINBL SUB-CIRCUITS																		

63A MCB; 300mA

		R	Y	B		
DB LOAD CABLE SIZE:	CONNECTED (W)	**2568.00**	**2330**	**2852.80**		*EARTH WIRE SIZE:*
4C x 10mm² PVC/PVC COPPER CABLE	SPARE(W)				**500**	**1 x 10mm² PVC SINGLE CORE COPPER CABLE**
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***8250.40***	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**10313**	VA
	CURRENT (A)				**14.4**	A

FIRST FLOOR DISTRIBUTION BOARD (DB-D)

Location:	PANEL ROOM	**Type:**	63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB-D
Cable Length(m) to ControlPanel		**Fed from**	CP1

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
DL1 - LIGHTING	2 x 2.5	20	100	2000	0.45	900.00			1	900.00			0.45	2000	100	20	2 x 2.5	DL2 - LIGHTING
DL3 - LIGHTING	2 x 2.5	20	100	2000	0.45		900.00		2		1125.00		0.45	2500	100	25	2 x 2.5	DL4 - LIGHTING
DL5 - LIGHTING	2 x 2.5	15	100	1500	0.45			675.00	3			720.00	0.45	1600	100	16	2 x 2.5	DL6 - LIGHTING
DL10 - LIGHTING	2 x 2.5	6	100	600	0.45	270.00			4	450			0.45	1000	1000	1	2 x 2.5	DL8 - LIGHTING
DL9 - LIGHTING	2 x 1.5	1	1000	1000	0.45		450		5		0		0.45	0	100	0	2 x 2.5	SPARE
DL11 - LIGHTING	2 x 2.5	22	100	2200	0.45			990.00	6			1035	0.45	2300	100	23	2 x 2.5	DL12 - LIGHTING
DL13 - LIGHTING	2 x 2.5	17	100	1700	0.45	765.00			7	450			0.45	1000	1000	1	2 x 2.5	DL14 - LIGHTING
DL7 - LIGHTING	2 x 2.5	27	100	2700	0.45		1215		8		0		0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	1000	0	0.45			0.00	9			0	0.45	0	1000	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45	0				0			0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45		0				0		0.4	0	6000	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45			0				0	0.5	0	1000	0	2 x 2.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				13700		1935	2565	1665		1800.00	1125	1755.00		10400				
* RING MAIN FINAL SUB-CIRCUITS																		

63A MCB; 300mA

		R	Y	B		
DB LOAD CABLE SIZE:	CONNECTED (W)	**3735.00**	**3690**	**3420.00**		*EARTH WIRE SIZE:*
4C x 6mm² PVC/PVC COPPER CABLE	SPARE(W)				**500**	**1 x 4mm² PVC SINGLE CORE COPPER CABLE**
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***11345.00***	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**14181.25**	VA
	CURRENT (A)				**19.8**	A

FIRST FLOOR DISTRIBUTION BOARD (DB-E)

Location:	PANEL ROOM	**Type:**	63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB-E
Cable Length(m) to ControlPanel		**Fed from**	CP1

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
ES1 - 13A SOCKET OUTLET	3 x 2.5	7	250	1750	0.50	875.00			1			1125.00	0.50	2250	250	9	3 x 2.5	ES2 - 13A SOCKET OUTLET
ES3 - 13A SOCKET OUTLET	3 x 2.5	8	250	2000	0.50		1000.00		2		1375.00		0.50	2750	250	11	3 x 2.5	ES4 - 13A SOCKET OUTLET
ES5 - 13A SOCKET OUTLET	3 x 2.5	8	250	2000	0.50			1000.00	3			375.00	0.50	750	250	3	3 x 2.5	ES6 - 13A SOCKET OUTLET
ES7 - 13A SOCKET OUTLET	3 x 2.5	8	250	2000	0.50	1000.00			4	750			0.50	1500	250	6	3 x 2.5	ES8 - 13A SOCKET OUTLET
ES9 - 13A SOCKET OUTLET	3 x 2.5	4	250	1000	0.50		500		5		1000		0.50	2000	250	8	3 x 2.5	ES10 - 13A SOCKET OUTLET
ES11 - 13A SOCKET OUTLET	3 x 2.5	13	250	3250	0.50			1625.00	6			625	0.50	1250	250	5	3 x 2.5	ES12 - 13A SOCKET OUTLET
SPARE	3 x 2.5	0	250	0	0.50	0.00			7	0			0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50		0		8		0		0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50			0.00	9			0	0.50	0	250	0	3 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45	0				0			0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 1.5	0	100	0	0.45		0				0		0.4	0	6000	0	3 x 4.0	SPARE
SPARE	2 x 2.5	0	100	0	0.45			0				0	0.5	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0				0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				12000		1875	1500	2625		1875.00	2375	1000.00		10500				
* RING MAIN FINAL SUB-CIRCUITS																		

63A MCB; 300mA

		R	Y	B		
DB LOAD CABLE SIZE:	CONNECTED (W)	**3750.00**	**3875**	**3625.00**		*EARTH WIRE SIZE:*
4C x 6mm² PVC/PVC COPPER CABLE	SPARE(W)				**500**	**1 x 4mm² PVC SINGLE CORE COPPER CABLE**
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***11750.00***	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**14687.5**	VA
	CURRENT (A)				**20.5**	A

FIRST FLOOR DISTRIBUTION BOARD (DB-G)

Location: PANEL ROOM — **Type:** 63A 8-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.: 230/400V, 50Hz — **DB Ref.:** DB-G
Cable Length(m) to ControlPanel — **Fed from** CP1

DESIGNATION OF LOAD	CABLE SIZE (mm²)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm²)	DESIGNATION OF LOAD
GAC1 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80	448.00			1	13	448.00			0.80	560	560	1	3 x 2.5	GAC2 - 20A SWITCH OUTLET
GAC3 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80		92.80		2	14		92.80		0.80	116	116	1	3 x 2.5	GAC5 - 20A SWITCH OUTLET
GAC5 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80			92.80	3	15			44.80	0.80	56	56	1	3 x 2.5	GAC4 - 20A SWITCH OUTLET
GAC7 - 20A SWITCH OUTLET	3 x 2.5	1	56	56	0.80	44.80			4	16	448			0.80	560	560	1	3 x 2.5	GAC9 - 20A SWITCH OUTLET
GAC11 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80		448.00		5	17		448.00		0.80	560	560	1	3 x 2.5	GAC10 - 20A SWITCH OUTLET
GAC9 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80			92.80	6	18			448	0.80	560	560	1	3 x 2.5	GAC12 - 20A SWITCH OUTLET
GAC13 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80	92.80			7	19	44.8			0.80	56	56	1	3 x 2.5	GAC14 - 20A SWITCH OUTLET
GAC15 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80		92.8		8	20		92.8		0.80	116	116	1	3 x 2.5	GAC16 - 20A SWITCH OUTLET
GAC17 - 20A SWITCH OUTLET	3 x 2.5	1	56	56	0.80			44.80	9	21			1080	0.80	1350	1350	1	3 x 4.0	GAC18 - 20A SWITCH OUTLET
G-ERV1 - ENERGY RECOVERY UNIT	3 x 2.5	1	400	400	0.80	320			10	22	0			0.80	0	1500	0	3 x 4.0	SPARE
G-ERV2 - ENERGY RECOVERY UNIT	3 x 2.5	1	400	400	0.80		320		11	23		0		0.80	0	1500	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	116	0	0.80			0	12	24			0	0.80	0	1500	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0					0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0					0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0					0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0					0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0					0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0					0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				2612		906	953.6	236.4			940.00	634	1572.80		3934				
* RING MAIN FINAL SUB-CIRCUITS									63A MCB, 300mA										

	R	Y	B			
DB LOAD CABLE SIZE:						*EARTH WIRE SIZE:*
4C x 6mm² PVC/PVC COPPER CABLE	CONNECTED (W)	**1046.40**	**1587.20**	**1803.20**		**1 x 6mm² PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				**5736.60**	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**7171**	VA
	CURRENT (A)				**10**	A

SECOND FLOOR DISTRIBUTION BOARD (DB-H)

Location: PANEL ROOM — **Type:** 63A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.: 230/400V, 50Hz — **DB Ref.:** DB-H
Cable Length(m) to ControlPanel — **Fed from** CP1

DESIGNATION OF LOAD	CABLE SIZE (mm²)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm²)	DESIGNATION OF LOAD
HL1 - LIGHTING	2 x 2.5	16	100	1600	0.45	720.00			1	10	720.00			0.45	1600	100	16	2 x 2.5	HL2 - LIGHTING
HL3 - LIGHTING	2 x 2.5	31	100	3100	0.45		1395.00		2	11		945.00		0.45	2100	100	21	2 x 2.5	HL4 - LIGHTING
HL5 - LIGHTING	2 x 2.5	15	100	1500	0.45			675.00	3	12			450.00	0.45	1000	100	1	2 x 1.5	HL6 - LIGHTING
SPARE	2 x 2.5	0	100	0	0.45	0.00			4	13	945			0.45	2100	100	21	2 x 2.5	HL8 - LIGHTING
HL9 - LIGHTING	2 x 2.5	17	100	1700	0.45		765		5	14		0		0.45	0	100	0	2 x 2.5	SPARE
HL11 - LIGHTING	2 x 1.5	1	1000	1000	0.45			450.00	6	15			450	0.45	1000	100	1	2 x 1.5	HL12 - LIGHTING
HL13 - LIGHTING	2 x 2.5	16	100	1600	0.45	720.00			7	16	0			0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45		0		8	17		0		0.45	0	100	0	2 x 2.5	SPARE
HL7 - LIGHTING	2 x 2.5	12	100	1200	0.45			540.00	9	18			585	0.45	1300	100	13	2 x 2.5	HL10 - LIGHTING
SPARE	2 x 2.5	0	100	0	0.45	0					0			0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45		0					0		0.4	0	6000	0	2 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45			0					0	0.5	0	1000	0	2 x 2.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0					0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0					0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0					0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0					0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0					0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0					0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				11700		1440	2160	1665			1665.00	945	1485.00		9100				
* RING MAIN FINAL SUB-CIRCUITS									63A MCB, 300mA										

	R	Y	B			
DB LOAD CABLE SIZE:						*EARTH WIRE SIZE:*
4C x 6mm² PVC/PVC COPPER CABLE	CONNECTED (W)	**3105.00**	**3105**	**3150.00**		**1 x 4mm² PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				**9060.00**	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**12325**	VA
	CURRENT (A)				**17.2**	A

SECOND FLOOR DISTRIBUTION BOARD (DB-J)

Location: PANEL ROOM — **Type:** 63A 4-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.: 230/400V, 50Hz — **DB Ref.:** DB-J
Cable Length(m) to ControlPanel — **Fed from** CP1

DESIGNATION OF LOAD	CABLE SIZE (mm²)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm²)	DESIGNATION OF LOAD
SPARE	3 x 2.5	0	250	0	0.50	0.00			1	7	1000.00			0.50	2000	250	8	3 x 2.5	JS2 - 13A SOCKET OUTLET
JS3 - 13A SOCKET OUTLET	3 x 2.5	11	250	2750	0.50		1375.00		2	8		1250.00		0.50	2500	250	10	3 x 2.5	JS4 - 13A SOCKET OUTLET
JS9 - 13A SOCKET OUTLET	3 x 2.5	4	250	1000	0.50			500.00	3	9			1000.00	0.50	2000	250	8	3 x 2.5	JS6 - 13A SOCKET OUTLET
JS7 - 13A SOCKET OUTLET	3 x 2.5	9	250	2250	0.50	1125.00			4	10	500			0.50	1000	250	4	3 x 2.5	JS8 - 13A SOCKET OUTLET
JS5 - 13A SOCKET OUTLET	3 x 2.5	3	250	750	0.50		375		5	11		0		0.50	0	250	0	3 x 2.5	SPARE
JS1 - 13A SOCKET OUTLET	3 x 2.5	11	250	2750	0.50			1375.00	6	12			0	0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50	0.00					0			0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50		0					0		0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	250	0	0.50			0.00					0	0.50	0	250	0	3 x 2.5	SPARE
SPARE	2 x 2.5	0	100	0	0.45	0					0			0.45	0	100	0	2 x 2.5	SPARE
SPARE	2 x 1.5	0	100	0	0.45		0					0		0.4	0	6000	0	3 x 4.0	SPARE
SPARE	2 x 2.5	0	100	0	0.45			0					0	0.5	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0					0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0					0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0					0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0					0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0					0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0					0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				9500		1125	1750	1875			1500.00	1250	1000.00		7500				
* RING MAIN FINAL SUB-CIRCUITS									63A MCB, 300mA										

	R	Y	B			
DB LOAD CABLE SIZE:						*EARTH WIRE SIZE:*
4C x 6mm² PVC/PVC COPPER CABLE	CONNECTED (W)	**2625.00**	**3000.00**	**2875.00**		**1 x 4mm² PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				**8000.00**	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**11250**	VA
	CURRENT (A)				**15.7**	A

SECOND FLOOR DISTRIBUTION BOARD (DB-K)

Location: PANEL ROOM — **Type:** 63A 8-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.: 230/400V, 50Hz — **DB Ref.:** DB-K
Cable Length(m) to ControlPanel — **Fed from** CP1

DESIGNATION OF LOAD	CABLE SIZE (mm²)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm²)	DESIGNATION OF LOAD
KAC1 - 20A SWITCH OUTLET	3 x 2.5	1	56	56	0.80	44.80			1	13	0.00			0.80	0	116	0	3 x 2.5	SPARE
KAC3 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80		92.80		2	14		92.80		0.80	116	116	1	3 x 2.5	KAC4 - 20A SWITCH OUTLET
SPARE	3 x 2.5	0	116	0	0.80			0.00	3	15			92.80	0.80	116	116	1	3 x 2.5	KAC2 - 20A SWITCH OUTLET
	3 x 2.5	0	560	0	0.80	0.00			4	16	0.00			0.80	0	116	0	3 x 2.5	SPARE
KAC9 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80		92.80		5	17		92.80		0.80	116	116	1	3 x 2.5	KAC10 - 20A SWITCH OUTLET
KAC11 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80			448.00	6	18			92.80	0.80	116	116	1	3 x 2.5	KAC8 - 20A SWITCH OUTLET
KAC15 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80	92.80			7	19	0.00			0.80	0	116	0	3 x 2.5	SPARE
KAC7 - 20A SWITCH OUTLET	3 x 2.5	1	560	560	0.80		448.00		8	20		92.80		0.80	116	116	1	3 x 4.0	KAC13 - 20A SWITCH OUTLET
K-ERV1 - ENERGY RECOVERY UNIT	3 x 2.5	1	400	400	0.80			320.00	9	21			92.80	0.80	116	116	1	3 x 4.0	KAC14 - 20A SWITCH OUTLET
K-ERV2 - ENERGY RECOVERY UNIT	3 x 2.5	1	400	400	0.80	320.00			10	22	680.00			0.80	850	850	1	3 x 4.0	KAC12 - 20A SWITCH OUTLET
KAC5 - 20A SWITCH OUTLET	3 x 2.5	1	116	116	0.80		92.80		11	23		92.80		0.80	116	116	1	3 x 2.5	KAC6 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	116	0	0.80			0.00	12	24			0.00	0.80	0	1500	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0.00					0.00			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0.00					0.00		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0.00					0.00	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0.00					0.00			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0.00					0.00		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0.00					0.00	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				2440		457.60	726.40	768.00			680.00	371.20	278.40		1662				
* RING MAIN FINAL SUB-CIRCUITS									63A MCB, 300mA										

	R	Y	B			
DB LOAD CABLE SIZE:						*EARTH WIRE SIZE:*
4C x 6mm² PVC/PVC COPPER CABLE	CONNECTED (W)	**1137.60**	**1097.60**	**1046.40**		**1 x 6mm² PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				**3781.60**	W
63A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**4727**	VA
	CURRENT (A)				**6.6**	A

Asset Metadata

Item	Description
Project ID	P003
Building Type	2-Storey Residential Complex
Engineering Discipline	Electrical Engineering
Asset Category	Lighting, Power & ELV Systems
Software	AutoCAD
File Formats	DWG
Drawing Status	Completed Design

Project Overview

This project involved the complete electrical and Extra Low Voltage (ELV) system design for a two-storey residential complex. The design integrates lighting, power distribution, fire alarm, CCTV, and load scheduling to provide a coordinated electrical installation suitable for residential occupancy.

Scope of Work

- Lighting layout design for all floors
- Power distribution layout design
- Fire alarm layout design
- Fire alarm schematic preparation
- CCTV system layout design
- Load calculation and schedule preparation
- Circuit distribution planning
- Consumer Unit (CU) arrangement

Engineering Assets Produced

- Lighting Layout Plans
- Power Layout Plans

- Fire Alarm Layout Plans
- CCTV Layout Plans
- Electrical Load Schedule
- Circuit Distribution Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout Plans	Lighting for 2-Storey Residential Complex	DWG
P-001	Power Layout Plans	Power for 2-Storey Residential Complex	DWG
ELV-001	Extra Low Voltage Plans	ELV for 2-Storey Residential Complex	DWG
LS-001	Load Schedule Plan	DB for 2-Storey Residential Complex	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Microsoft Excel Load Schedule (.xlsx)

Design Considerations

The lighting system was designed to provide balanced illumination, efficient fixture placement, and logical circuit grouping throughout the building.

The power distribution system was planned to ensure safe circuit loading, balanced distribution, appropriate protective device selection, and efficient cable routing.

The ELV systems were coordinated to provide effective fire detection, alarm notification, and CCTV surveillance with organized cable pathways and maintainable system layouts.

Electrical load calculations were completed to establish connected load, estimated demand load, circuit breaker selection, cable sizing basis, and circuit load balancing.

Repository Structure

P003_Residential_Complex/

├── Load Schedules

----- CAD_Drawings

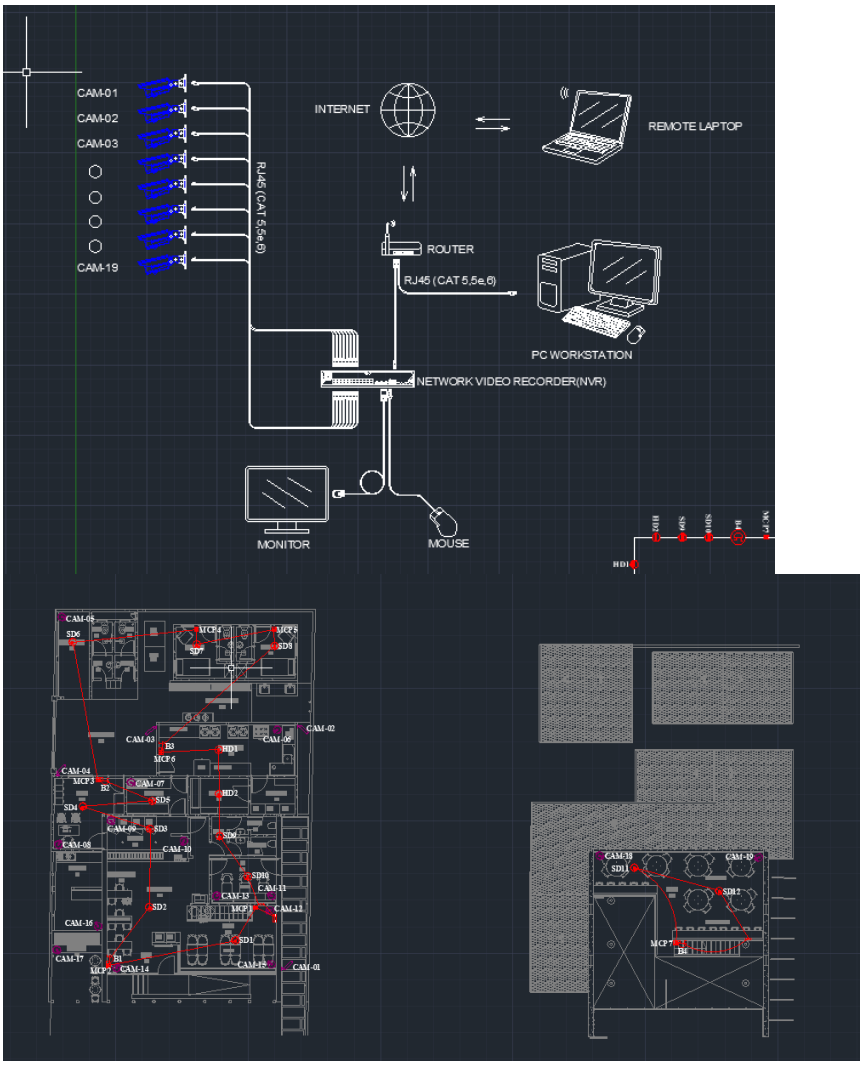
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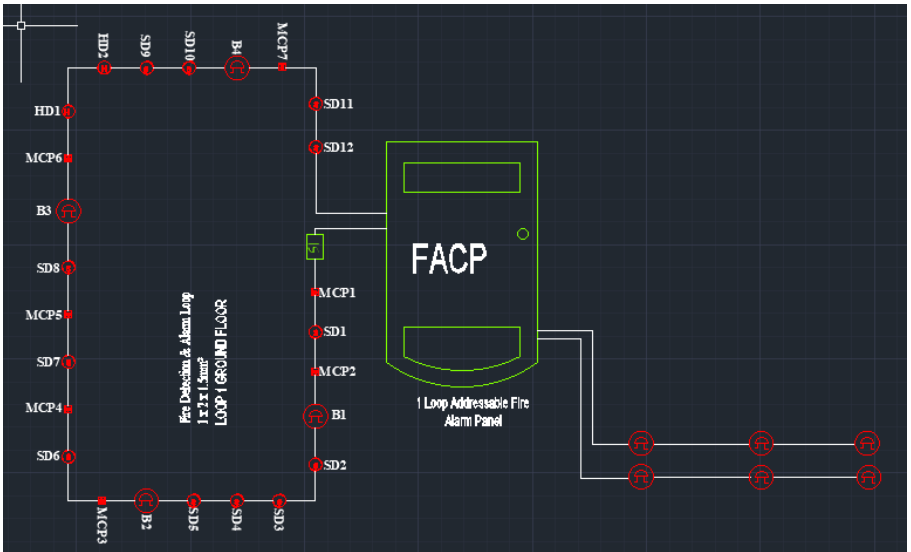
Skills Demonstrated

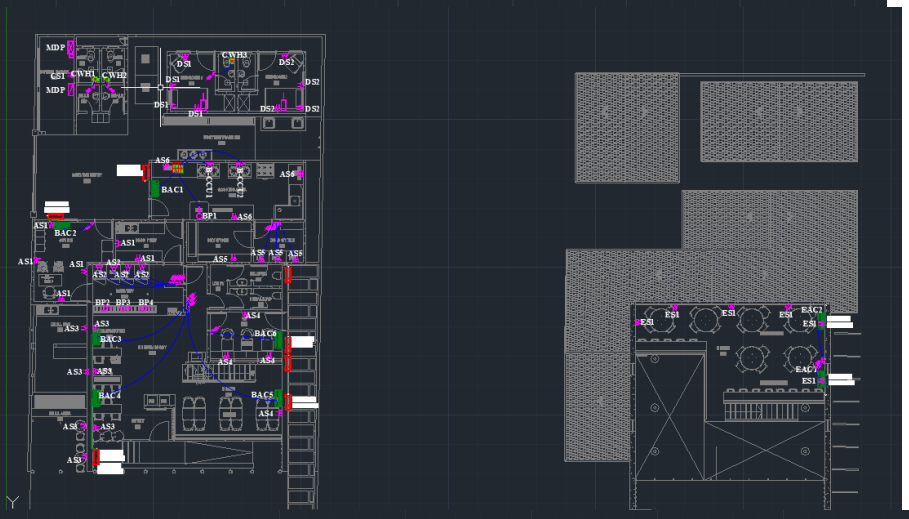
- Integrated electrical system design
- Residential power distribution planning
- ELV system coordination
- Fire alarm schematic preparation
- CCTV layout planning
- Load calculation and circuit balancing
- Professional engineering documentation

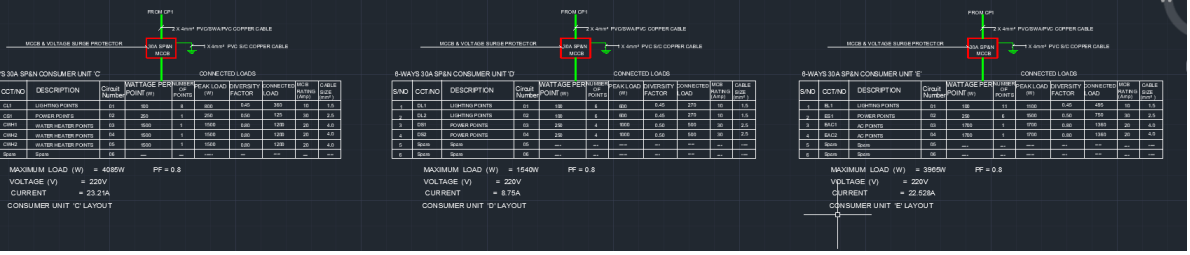
PROJECT 4

Lighting, Power, Fire Alarm and CCTV Design – Commercial Complex









6-WAYS 30A SPAN CONSUMER UNIT 'C'

S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
1	CU1	LIGHTING POINTS	01	180	9	800	0.45	360	10	1.5
2	CU2	POWER POINTS	02	250	1	250	0.80	1250	25	2.5
3	CWH1	WATER HEATER POINTS	03	1500	1	1500	0.80	1200	25	4.0
4	CWH2	WATER HEATER POINTS	04	1500	1	1500	0.80	1200	25	4.0
5	CWH3	WATER HEATER POINTS	05	1500	1	1500	0.80	1200	25	4.0
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 4085W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 23.21A
CONSUMER UNIT 'C' LAYOUT

6-WAYS 30A SPAN CONSUMER UNIT 'D'

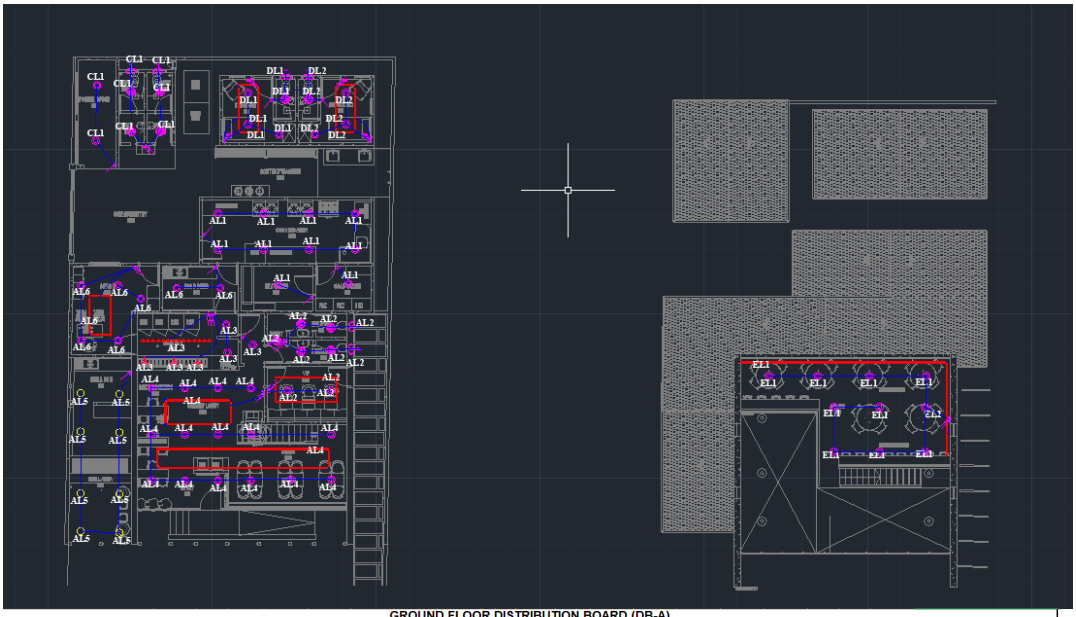
S/NO	CCT.NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
1	DL1	LIGHTING POINTS	01	150	6	900	0.45	270	10	1.5
2	DL2	LIGHTING POINTS	02	150	6	900	0.45	270	10	1.5
3	DB1	POWER POINTS	03	250	4	1000	0.50	500	30	2.5
4	DB2	POWER POINTS	04	250	4	1000	0.50	500	30	2.5
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 1540W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 8.75A
CONSUMER UNIT 'D' LAYOUT

6-WAYS 30A SPAN CONSUMER UNIT 'E'

S/NO	CCT.NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
1	EL1	LIGHTING POINTS	01	150	11	1100	0.45	495	10	1.5
2	ES1	POWER POINTS	02	250	6	1500	0.50	750	30	2.5
3	EAC1	AC POINTS	03	1750	1	1750	0.80	1365	20	4.0
4	EAC2	AC POINTS	04	1750	1	1750	0.80	1365	20	4.0
5	Spare	Spare	05	---	---	---	---	---	---	---
6	Spare	Spare	06	---	---	---	---	---	---	---

MAXIMUM LOAD (W) = 3865W PF = 0.8
VOLTAGE (V) = 220V
CURRENT = 22.528A
CONSUMER UNIT 'E' LAYOUT



GROUND FLOOR DISTRIBUTION BOARD (DB-A)

Location:	POWER HOUSE	**Type:**	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB-A
Cable Length(m) to ControlPanel		**Fed from**	MDP

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B		TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
AL5 - LIGHTING	2 x 1.5	8	100	800	0.45	360.00			1	16	10	450			0.45	1000	100	10	2 x 1.5	AL2 - LIGHTING
AL3 - LIGHTING	2 x 1.5	7	100	700	0.45		315.00		2	16	11		360.00		0.45	800	100	8	2 x 1.5	AL6 - LIGHTING
AL1 - LIGHTING	2 x 1.5	10	100	1000	0.45			450.00	3	16	12			765.00	0.45	1700	100	17	2 x 1.5	AL4 - LIGHTING
AS1 - 13A SOCKET OUTLET	3 x 2.5	6	250	1500	0.50	750			4	16	13	500			0.50	1000	250	4	3 x 2.5	AS2 - 13A SOCKET OUTLET
AS3 - 13A SOCKET OUTLET	3 x 2.5	7	250	1750	0.50		875.00		5	16	14		500		0.50	1000	250	4	3 x 2.5	AS4 - 13A SOCKET OUTLET
AS5 - 13A SOCKET OUTLET	3 x 2.5	4	250	1000	0.50			500.00	6	16	15			375	0.50	750	250	3	3 x 2.5	AS6 - 13A SOCKET OUTLET
SPARE	2 x 1.5	0	100	0	0.45	0			7	16	16	0			0.5	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.8		0		8	16	17		0		0.4	0	6000	0	3 x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.50			0.00	9	16	18			0	0.5	0	1000	0	2 x 1.5	SPARE
SPARE	2 x 1.5	0	100	0	0.45	0				16		0			0.5	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.8		0			16			0		0.4	0	6000	0	3 x 4.0	SPARE
SPARE	2 x 1.5	0	1000	0	0.5			0		16				0	0.5	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				16		0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0			16			0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0		16				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				16		0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0			16			0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0		16				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				6750		1110	1190	950				950.00	860	1140.00		6250				
* RING MAIN FINAL SUB-CIRCUITS										100A MCB, 300mA										

DB LOAD CABLE SIZE:		R	Y	B			*EARTH WIRE SIZE:*
4C x 4mm^2 PVC/PVC COPPER CABLE	CONNECTED (W)	**2060.00**	**2050**	**2090.00**			**1 x 4mm^2 PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**		
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***6700.00***	W	
100A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ	
	TOTAL CONNECTED (VA)				**8375**	VA	
	CURRENT (A)				**11.7**	A	

GROUND FLOOR DISTRIBUTION BOARD (DB-B)

Location:	POWER HOUSE	**Type:**	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB-B
Cable Length(m) to ControlPanel		**Fed from**	MDP

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B		TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
BAC1 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80	1360.00			1	16	10	1200			0.40	3000	3000	1	3 x 6.0	B-CCU2 - COOKER UNIT
BAC3 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80		1360.00		2	16	11		1360.00		0.80	1700	1700	1	3 x 4.0	BAC4 - 20A SWITCH OUTLET
BAC5 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80			1360.00	3	16	12			680.00	0.80	850	850	1	3 x 4.0	BAC6 - 20A SWITCH OUTLET
BP1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40	800.00			4	16	13	800			0.40	2000	2000	1	3 x 4.0	BP2 - 15A POWER SOCKET
BP3 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		5	16	14		800		0.40	2000	2000	1	3 x 4.0	BP4 - 15A POWER SOCKET
B-CCU1 - COOKER UNIT	3 x 6.0	1	3000	3000	0.40			1200.00	6	16	15			904.00	0.80	1130	1130	1	3 x 4.0	BAC2 - 20A SWITCH OUTLET
SPARE	3 x 4.0	0	2000	0	0.40	0.00			7	16	16	0			0.80	0	1500	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0		8	16	17		0		0.40	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0	9	16	18			0	1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.50	0				16		0			0.5	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.8		0			16			0		0.4	0	6000	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.5			0		16				0	0.7	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.8	0				16		0			0.6	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.6		0			16			0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0		16				0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0	0				16		0			1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0		0			16			0		1.0	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.0			0		16				0	1.0	0	400	0	3x 4.0	SPARE
SUB-TOTALS				12100		2160	2160	2560				2000.00	2160	1584.00		10680				
* RING MAIN FINBL SUB-CIRCUITS										100A MCB, 300mA										

DB LOAD CABLE SIZE:		R	Y	B			*EARTH WIRE SIZE:*
4C x 6mm^2 PVC/PVC COPPER CABLE	CONNECTED (W)	**4160.00**	**4320**	**4144.00**			**1 x 6mm^2 PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**		
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***13124.00***	W	
100A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ	
	TOTAL CONNECTED (VA)				**16405**	VA	
	CURRENT (A)				**22.9**	A	

Asset Metadata

Item	Description
Project ID	P004
Building Type	Commercial Complex
Engineering Discipline	Electrical Engineering
Asset Category	Power Distribution & ELV Systems
Software	AutoCAD, Microsoft Excel
File Formats	DWG, XLSX
Drawing Status	Completed Design

Project Overview

This project presents the electrical power distribution and Extra Low Voltage (ELV) system design for a commercial complex. The design includes power layouts, fire alarm systems, CCTV layouts, and electrical load documentation to support safe and reliable building operation.

Scope of Work

- Power layout design
- Lighting layout design
- Circuit distribution planning
- Fire alarm layout preparation
- Fire alarm schematic development
- CCTV layout design
- Load calculation
- Load schedule preparation
- Consumer Unit configuration

Engineering Assets Produced

- Power Layout Plans

- Lighting layout plans
- Circuit Distribution Drawings
- Fire Alarm Layout Plans
- Fire Alarm System Schematics
- CCTV Layout Plans
- Electrical Load Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout Plans	Lighting for Commercial Complex	DWG
P-001	Power Layout Plans	Power for Commercial Complex	DWG
SCH-001	Fire Alarm And CCTV Schematic	ELV for Commercial Complex	DWG
LS-001	Load Schedule	DB for Commercial Complex	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Microsoft Excel Load Schedule (.xlsx)

Design Considerations

The electrical power distribution was planned to provide balanced circuit loading, organised feeder routing, appropriate protective device selection, and efficient equipment supply arrangements.

The fire alarm system incorporates detector positioning, manual call point placement, system zoning, and notification device coverage to support reliable detection and maintenance.

The CCTV layout provides surveillance coverage of key circulation areas, entrances, and operational spaces with structured cable routing for installation and future servicing.

Electrical load calculations established connected load, demand load estimation, protective device sizing, and circuit distribution requirements.

Repository Structure

P004_Commercial_Complex/

├── Load_Schedules

├── CAD_Drawings

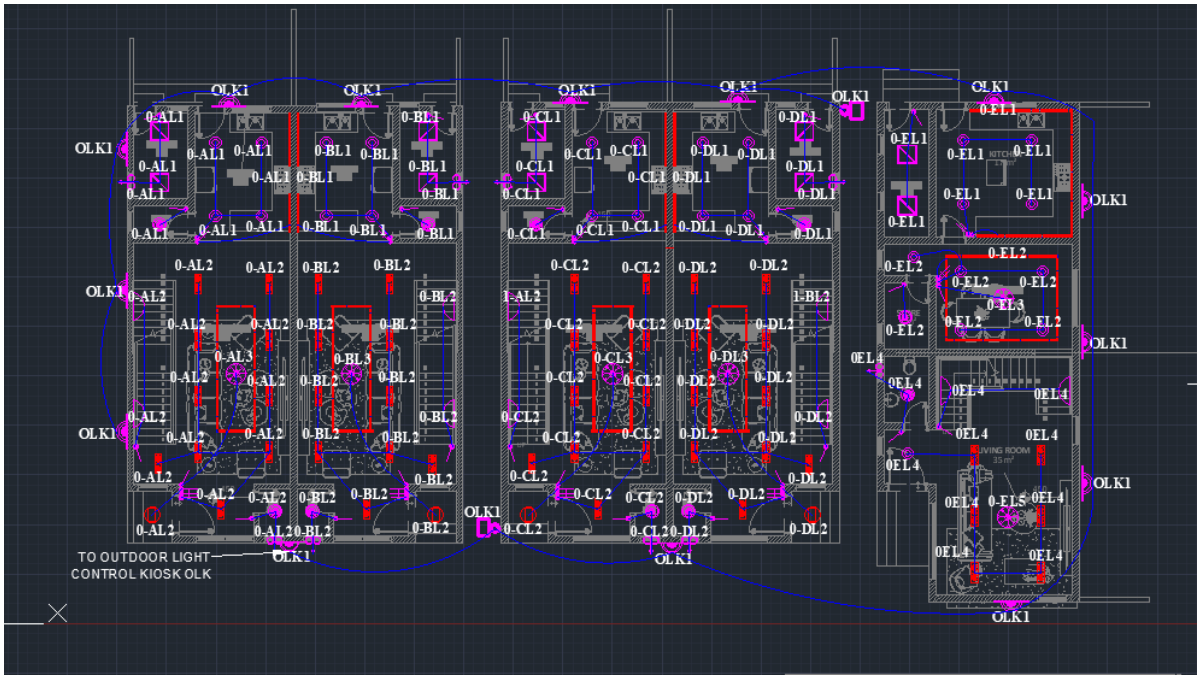
└── README.md

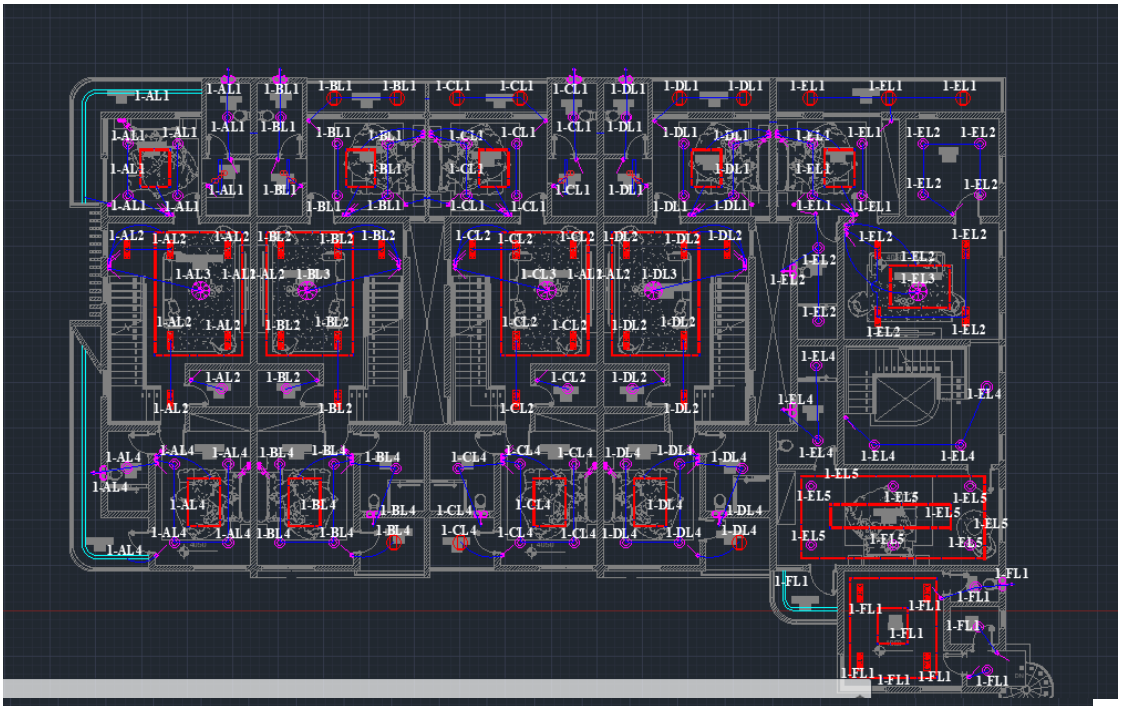
Skills Demonstrated

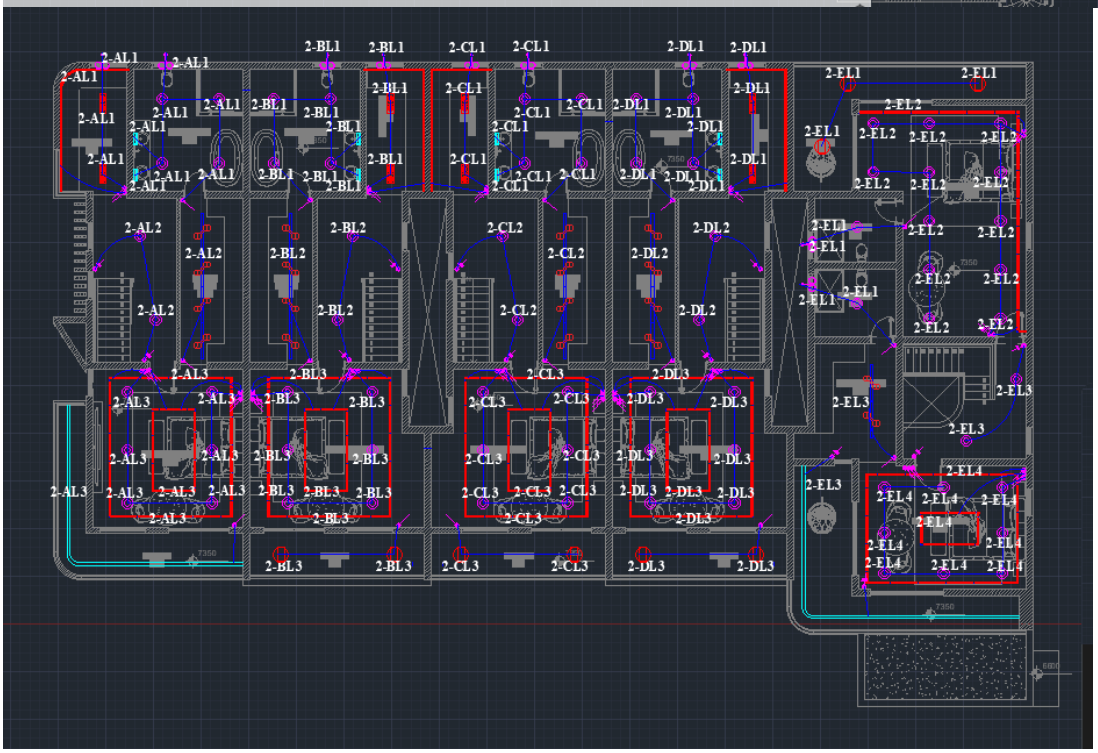
- Commercial electrical drafting
- Power distribution planning
- ELV system integration
- Fire alarm schematic preparation
- CCTV system planning
- Load analysis
- Engineering documentation

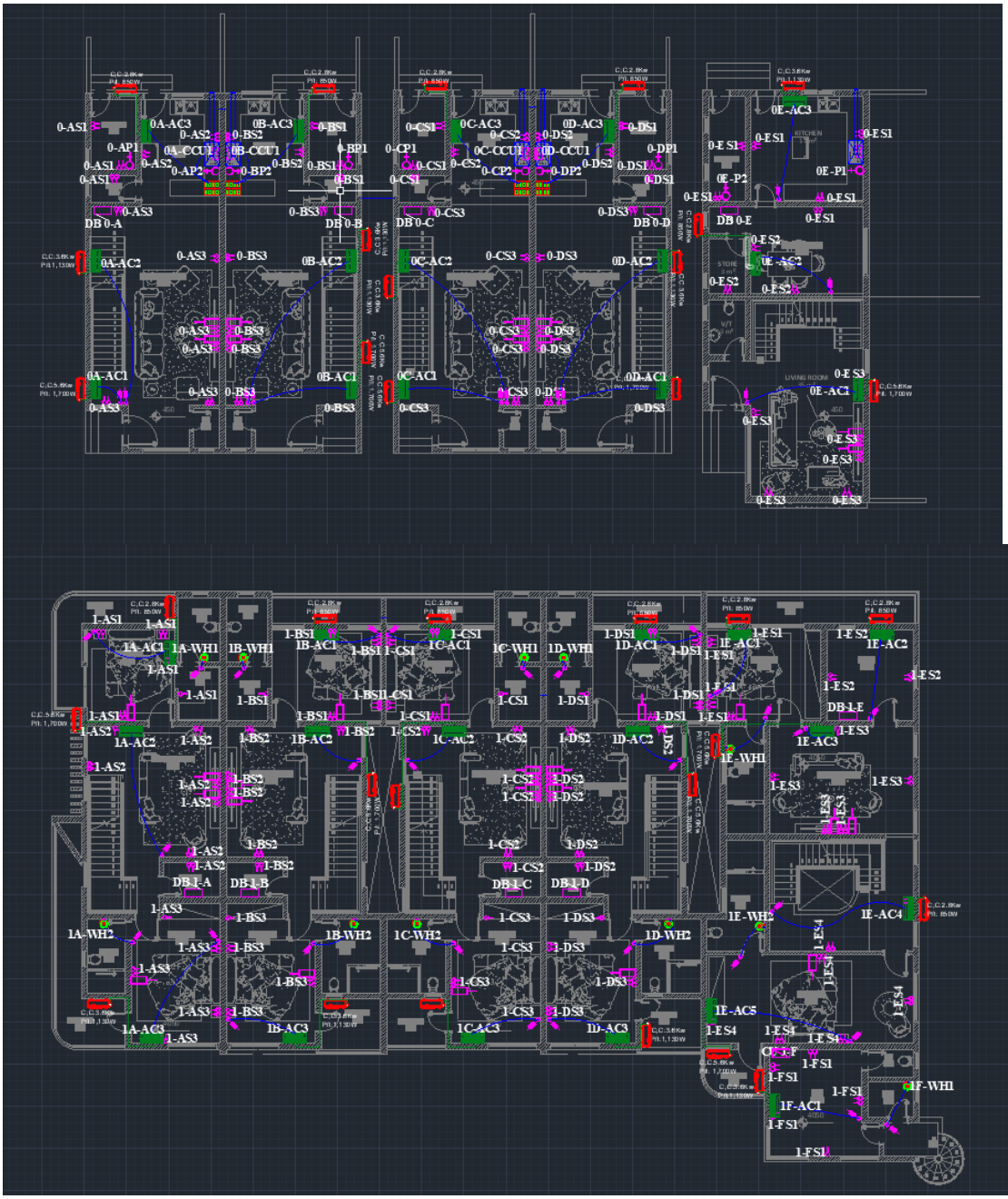
PROJECT 5

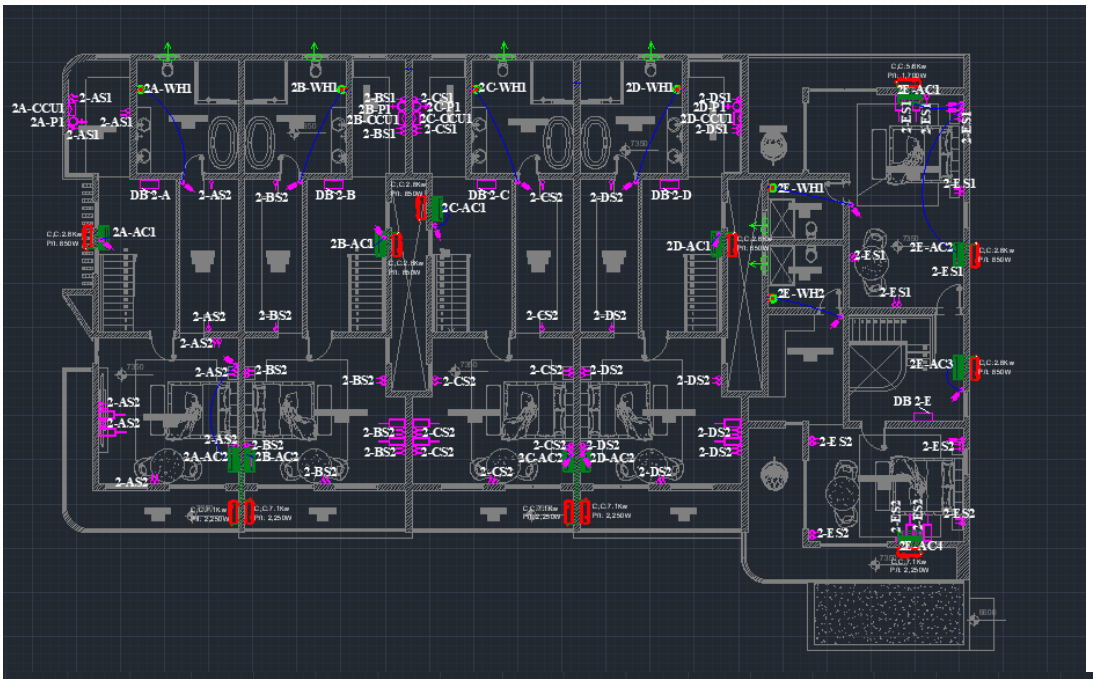
TAN Residence – Power, Lighting, Fire Alarm, CCTV and Consumer Unit Design

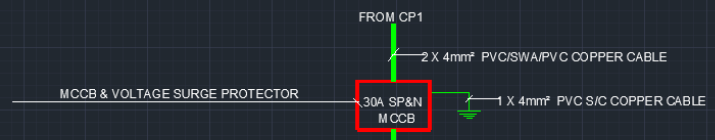



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1-F S1
1-F S1
1F-WH1
1F-AC1
1-F S1
1-F S1





6-WAYS 30A SP&N CONSUMER UNIT '1-F'

CONNECTED LOADS

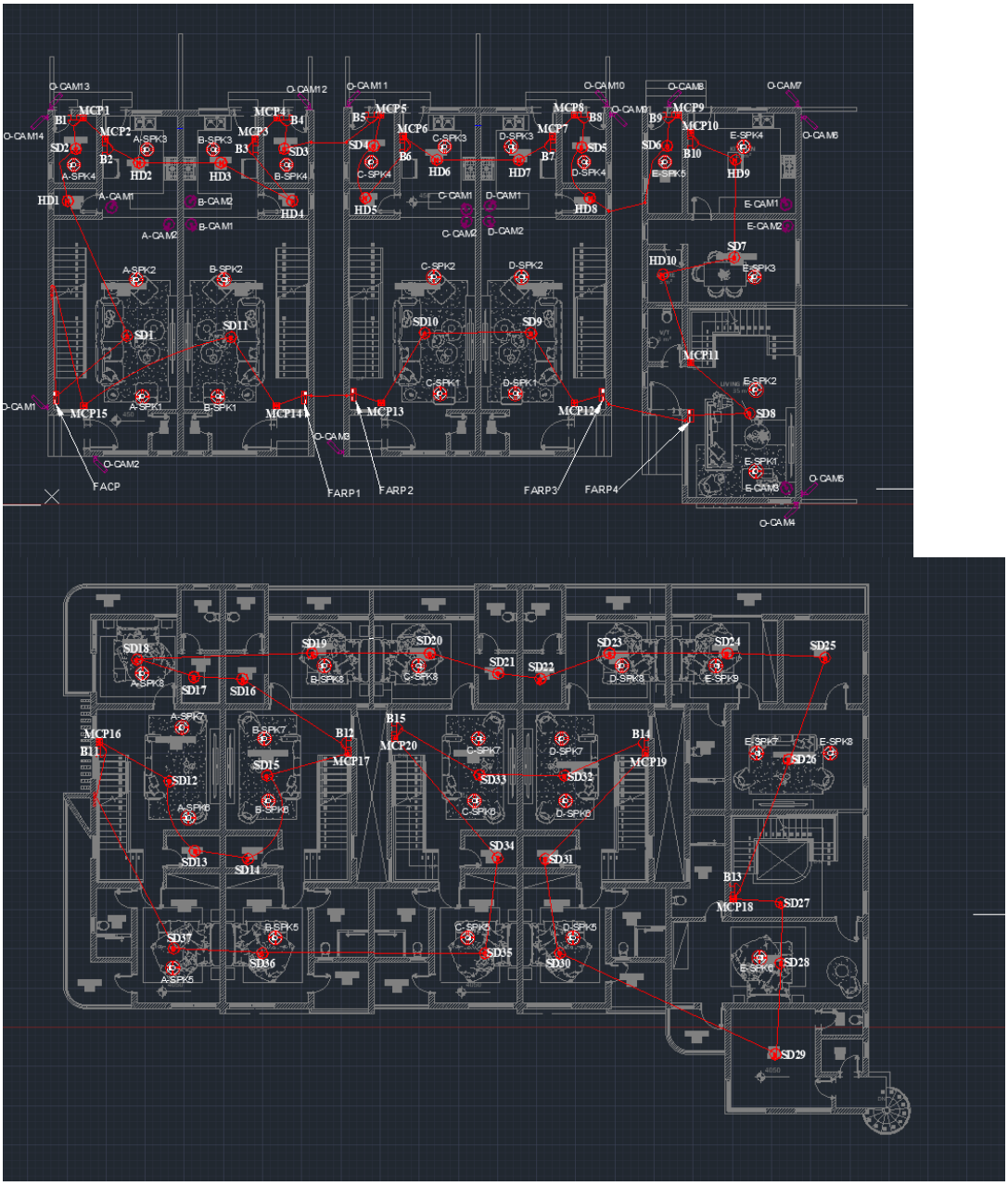
S/NO	CCT/NO	DESCRIPTION	Circuit Number	WATTAGE PER POINT (W)	NUMBER OF POINTS	PEAK LOAD (W)	DIVERSITY FACTOR	CONNECTED LOAD	MCB RATING (Amp)	CABLE SIZE (mm²)
1	1-FL1	LIGHTING POINTS	01	100	11	1100	0.45	495	10	1.5
2	1-FS1	POWER POINTS	02	250	5	1250	0.50	625	30	2.5
3	1F-AC1	AC POINTS	03	1130	1	1130	0.80	904	20	4.0
4	1F-WH1	WATER HEATER POINTS	04	1500	1	1500	0.80	1200	20	4.0
5	Spare	Spare	03	----	---	-----	---	----	---	----
6	Spare	Spare	03	----	---	-----	---	----	---	----

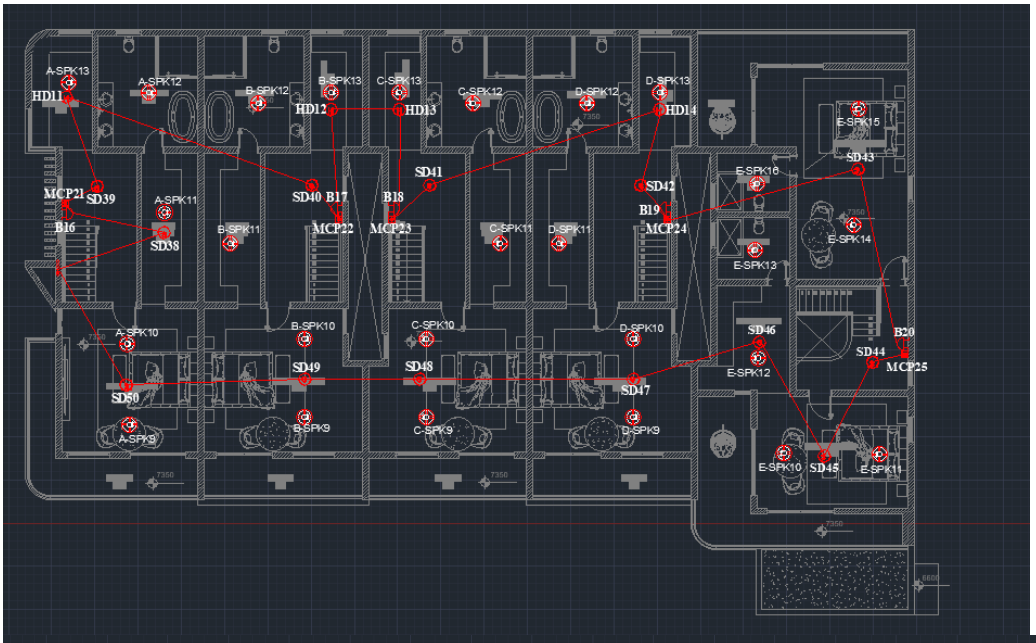
MAXIMUM LOAD (W) = 3224W PF = 0.8

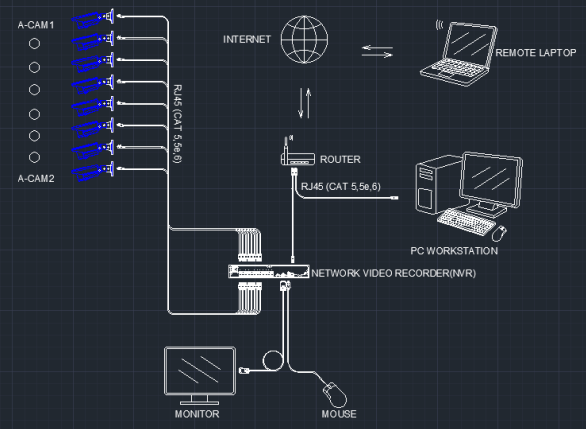
VOLTAGE (V) = 220V

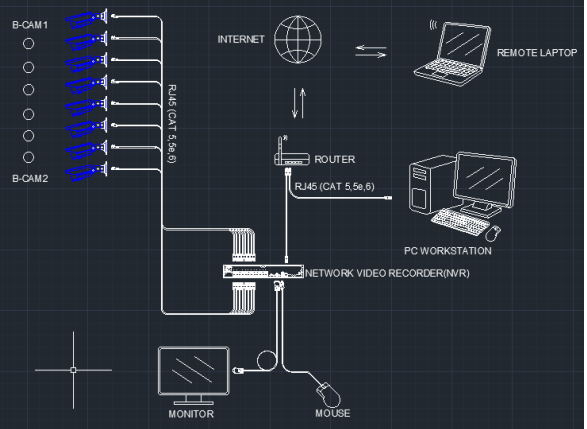
CURRENT = 18.318A

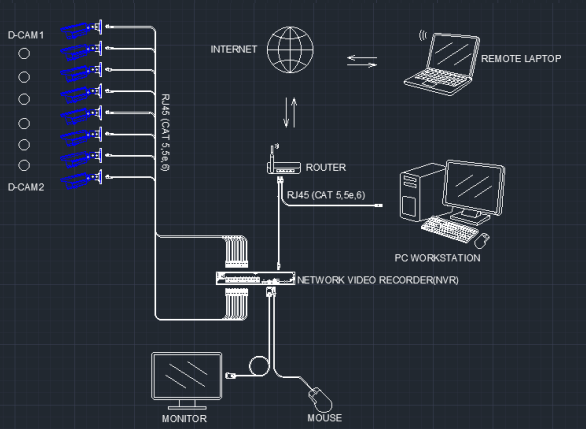
CONSUMER UNIT '1-F' LAYOUT

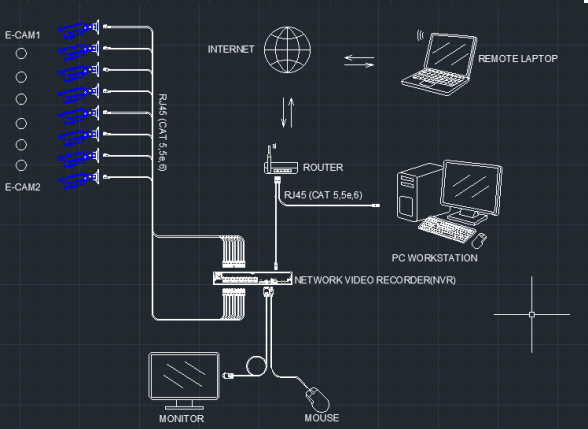

O-CAM13
O-CAM12
O-CAM11
O-CAM10
O-CAM9
O-CAM8
O-CAM7
O-CAM14
O-CAM6
MCP1
MCP2
MCP3
MCP4
MCP5
MCP6
MCP7
MCP8
MCP9
MCP10
B1
B2
B3
B4
B5
B6
B7
B8
B9
B10
SD1
SD2
SD3
SD4
SD5
SD6
SD7
SD8
SD9
SD10
SD11
HD1
HD2
HD3
HD4
HD5
HD6
HD7
HD8
HD9
HD10
A-SPK1
A-SPK2
A-SPK3
A-SPK4
B-SPK1
B-SPK2
B-SPK3
B-SPK4
C-SPK1
C-SPK2
C-SPK3
C-SPK4
D-SPK1
D-SPK2
D-SPK3
D-SPK4
E-SPK1
E-SPK2
E-SPK3
E-SPK4
E-SPK5
A-CAM1
A-CAM2
B-CAM1
B-CAM2
C-CAM1
C-CAM2
D-CAM1
D-CAM2
E-CAM1
E-CAM2
E-CAM3
MCP11
MCP12
MCP13
MCP14
MCP15
FACP
FARP1
FARP2
FARP3
FARP4
SD18
SD19
SD20
SD21
SD22
SD23
SD24
SD25
SD26
SD27
SD28
SD29
SD30
SD31
SD32
SD33
SD34
SD35
SD36
SD37
MCP16
MCP17
MCP18
MCP19
MCP20
B11
B12
B13
B14
B15
A-SPK6
A-SPK7
A-SPK8
B-SPK6
B-SPK7
B-SPK8
C-SPK6
C-SPK7
C-SPK8
D-SPK6
D-SPK7
D-SPK8
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E-SPK8
E-SPK9
A-SPK5
B-SPK5
C-SPK5
D-SPK5
E-SPK10

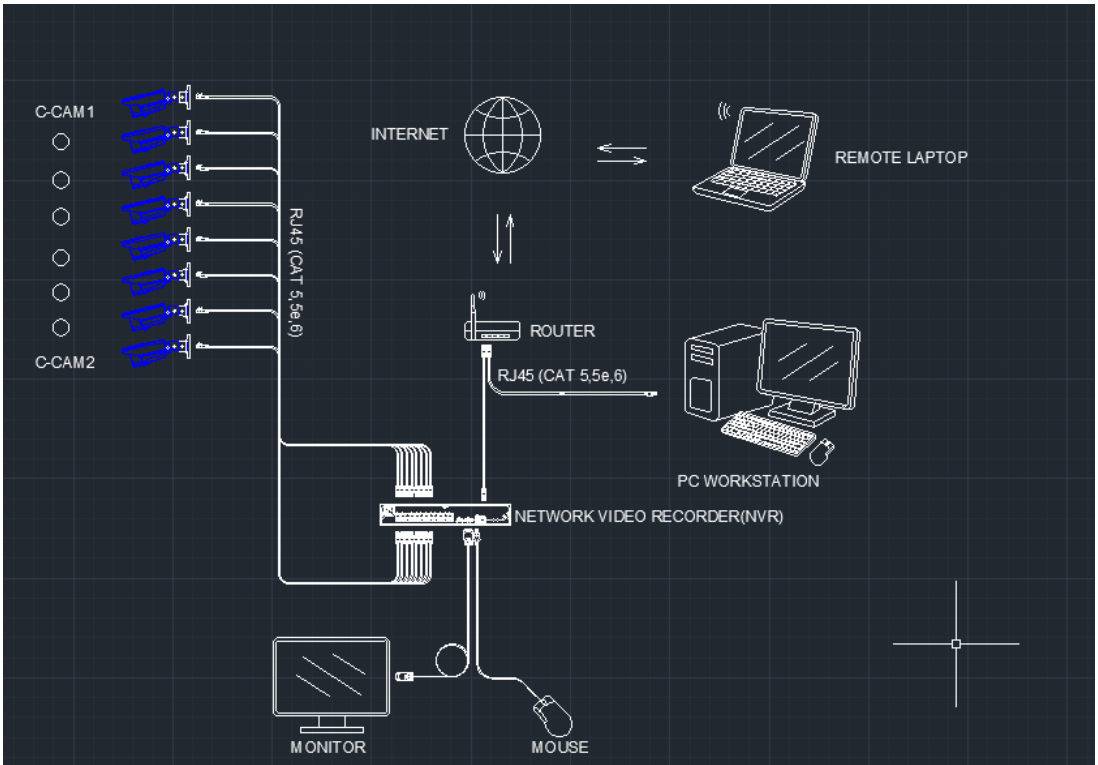

A-SPK13
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B-SPK12
B-SPK13
C-SPK13
C-SPK12
D-SPK12
D-SPK13
HD11
HD12
HD13
HD14
E-SPK15
SD43
E-SPK16
SD41
SD42
MCP21
SD39
A-SPK11
SD40
B17
B18
B19
B16
SD38
B-SPK11
MCP22
MCP23
C-SPK11
D-SPK11
MCP24
E-SPK14
E-SPK13
A-SPK10
B-SPK10
C-SPK10
D-SPK10
SD46
B20
SD44
MCP25
SD49
SD48
SD47
E-SPK12
SD50
A-SPK9
B-SPK9
C-SPK9
D-SPK9
E-SPK10
SD45
E-SPK11

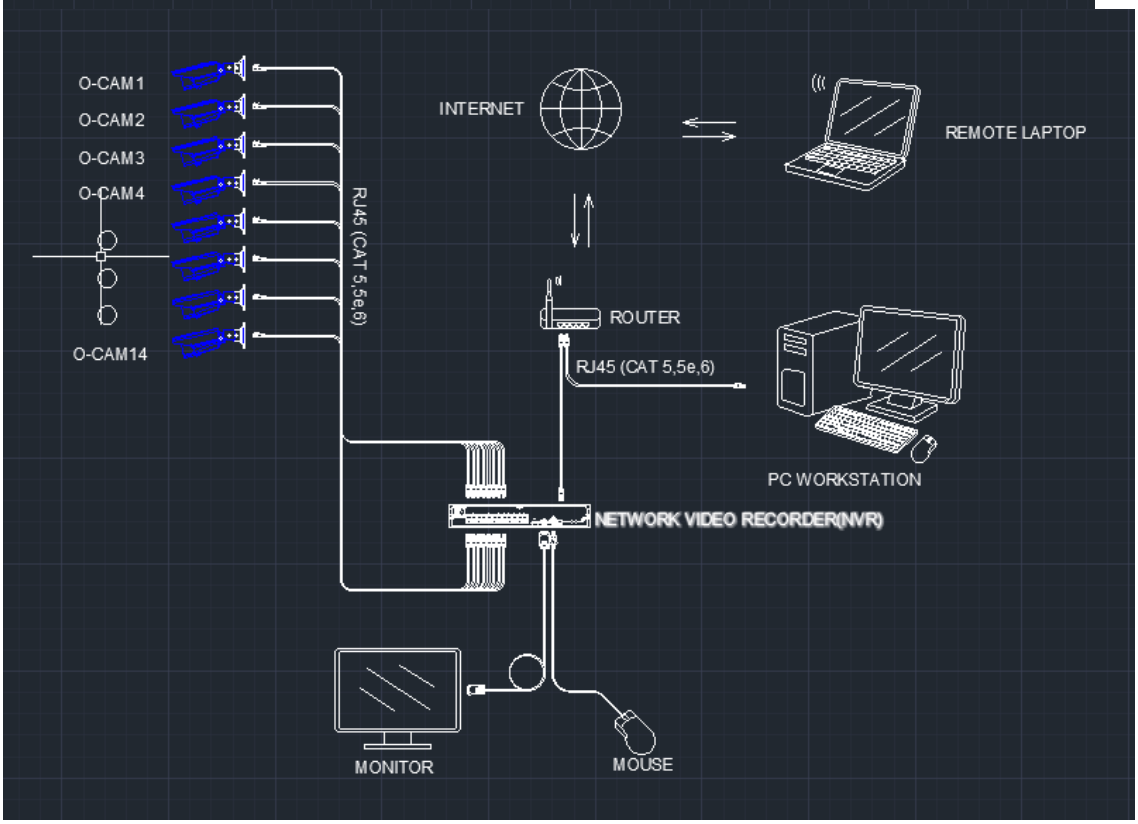

A-CAM1
A-CAM2
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE

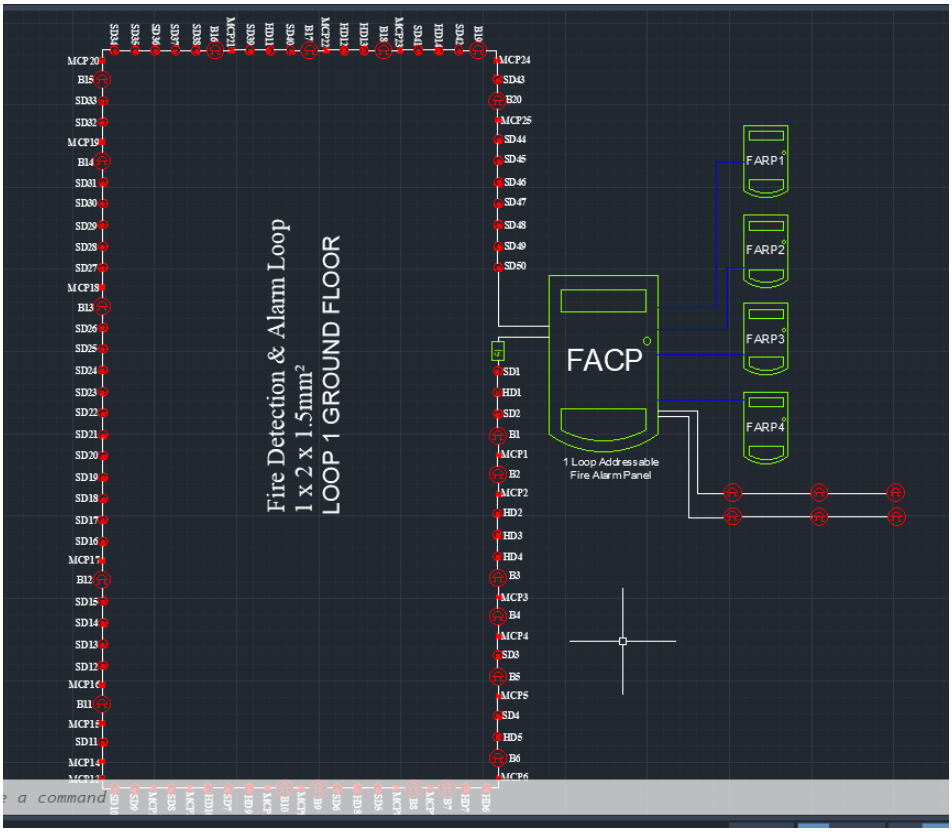

B-CAM1
B-CAM2
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE


D-CAM1
D-CAM2
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE


E-CAM1
E-CAM2
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE


C-CAM1
C-CAM2
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE


O-CAM1
O-CAM2
O-CAM3
O-CAM4
O-CAM14
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE



Asset Metadata

Item	Description
Project ID	P005
Building Type	Residential Building
Engineering Discipline	Electrical Engineering
Asset Category	Lighting, Power & ELV Systems
Software	AutoCAD
File Formats	DWG
Drawing Status	Completed Design

Project Overview

This project covers the electrical power, lighting, and Extra Low Voltage (ELV) system design for the TAN Residence. The design integrates lighting layouts, power distribution, fire alarm systems, CCTV layouts, and Consumer Unit arrangements into a coordinated residential electrical design package.

Scope of Work

- Lighting layout preparation
- Power distribution layout design
- Fire alarm layout preparation
- CCTV layout design
- Consumer Unit configuration

Engineering Assets Produced

- Lighting Layout Plans
- Power Layout Plans
- Fire Alarm Layout Plans
- CCTV Layout Plans
- Consumer Unit Arrangement

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout	Lighting for TAN Residence	DWG
P-001	Power Layout	Power for TAN Residence	DWG
SCH-001	Fire Alarm And CCTV Shematics	ELV for TAN Residence	DWG

Deliverables

- **AutoCAD Drawings (.dwg)**

Design Considerations

The lighting layout was developed to provide balanced illumination and practical switching arrangements throughout the residence.

The electrical power system provides organised socket outlet distribution, dedicated equipment supply points, and a structured Consumer Unit arrangement for safe circuit management.

The fire alarm and CCTV layouts were coordinated to provide effective safety monitoring, surveillance coverage, and maintainable cable routing throughout the property.

Repository Structure

P005_TAN_Residence/

├── CAD_Drawings

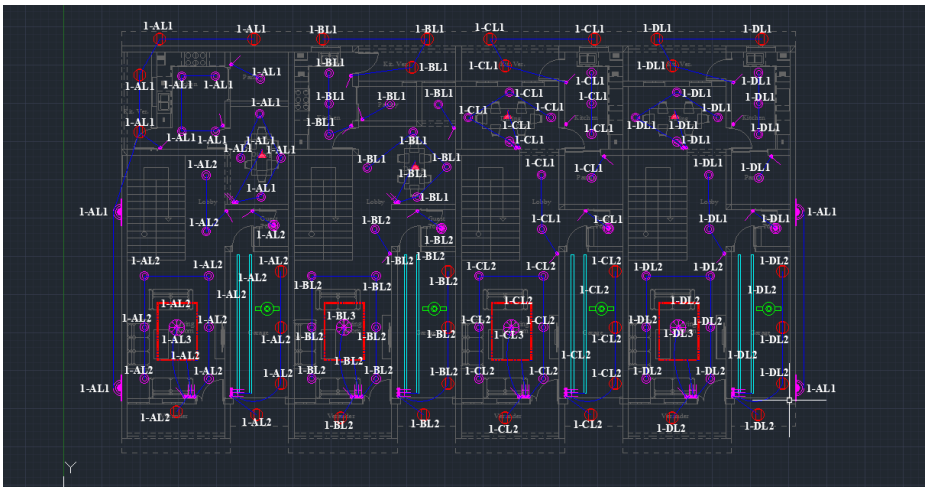
└── README.md

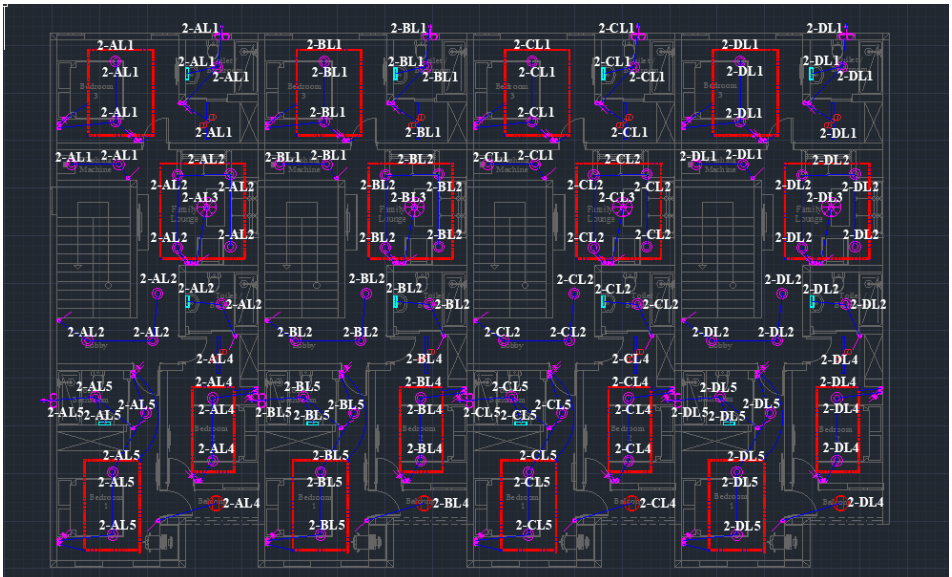
Skills Demonstrated

- Residential electrical design
- Consumer Unit configuration
- Lighting layout planning
- ELV system coordination
- CCTV layout preparation
- Fire alarm system planning
- Professional drafting

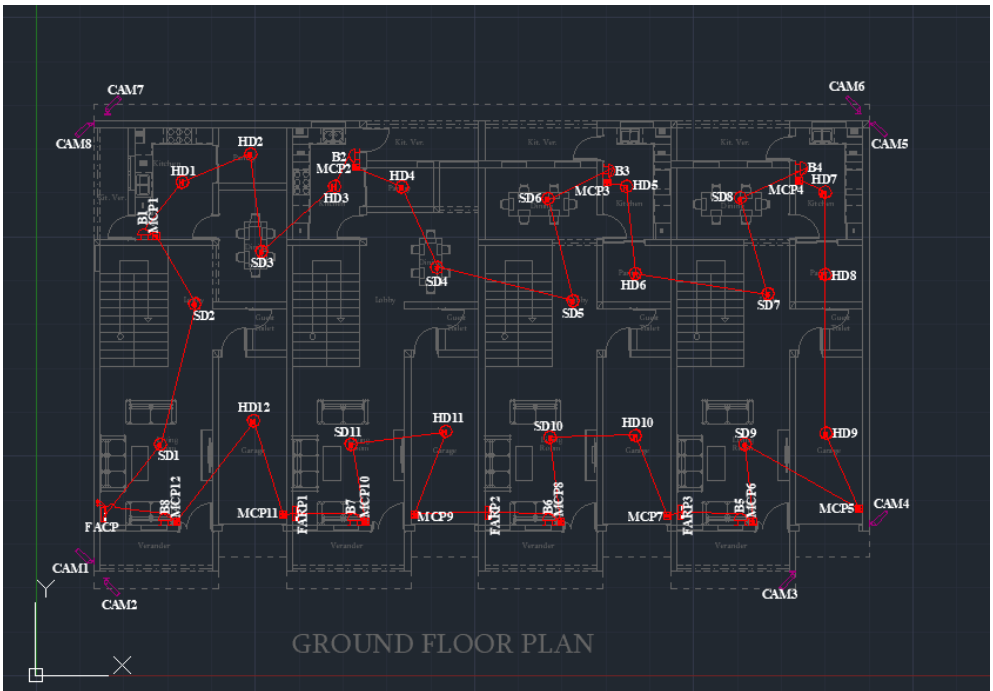
PROJECT 6

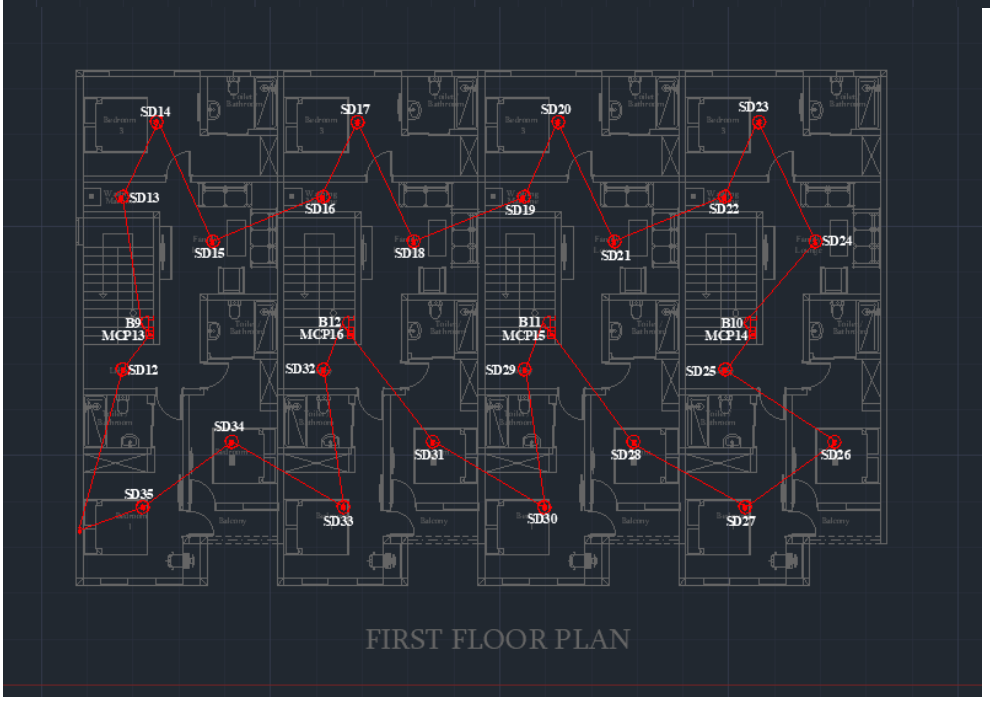
Terrace Building – Power, Lighting, Fire Alarm and CCTV Design

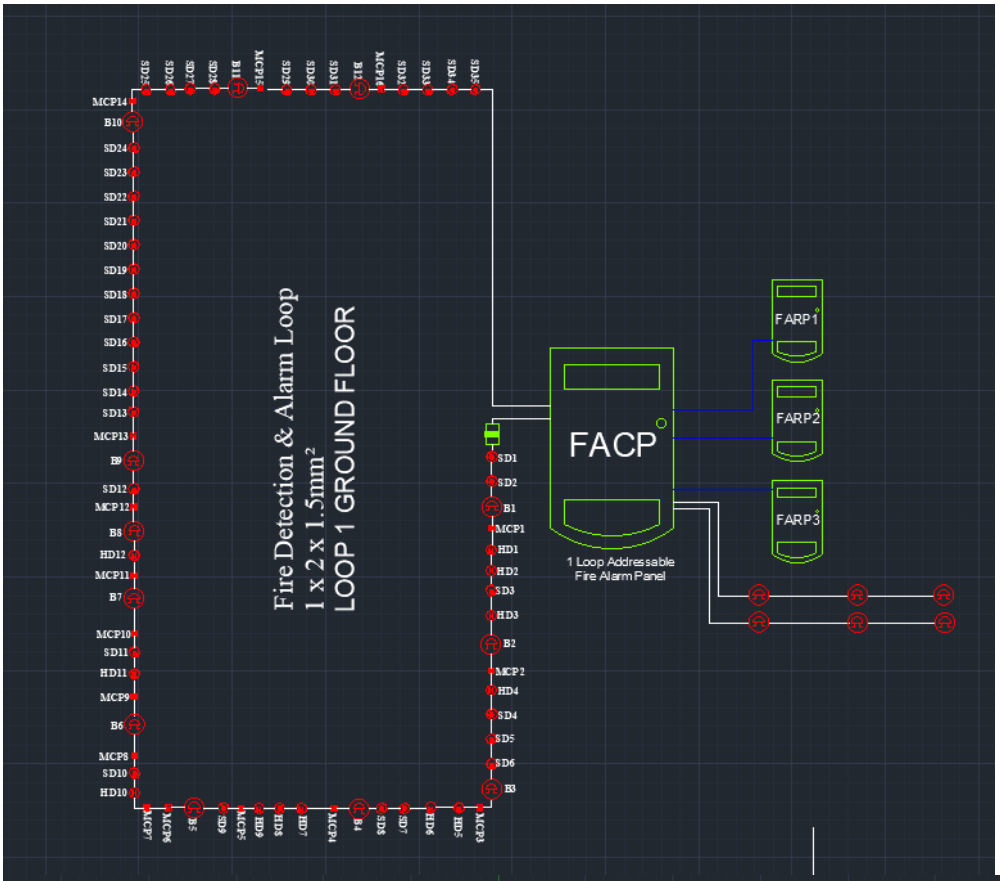


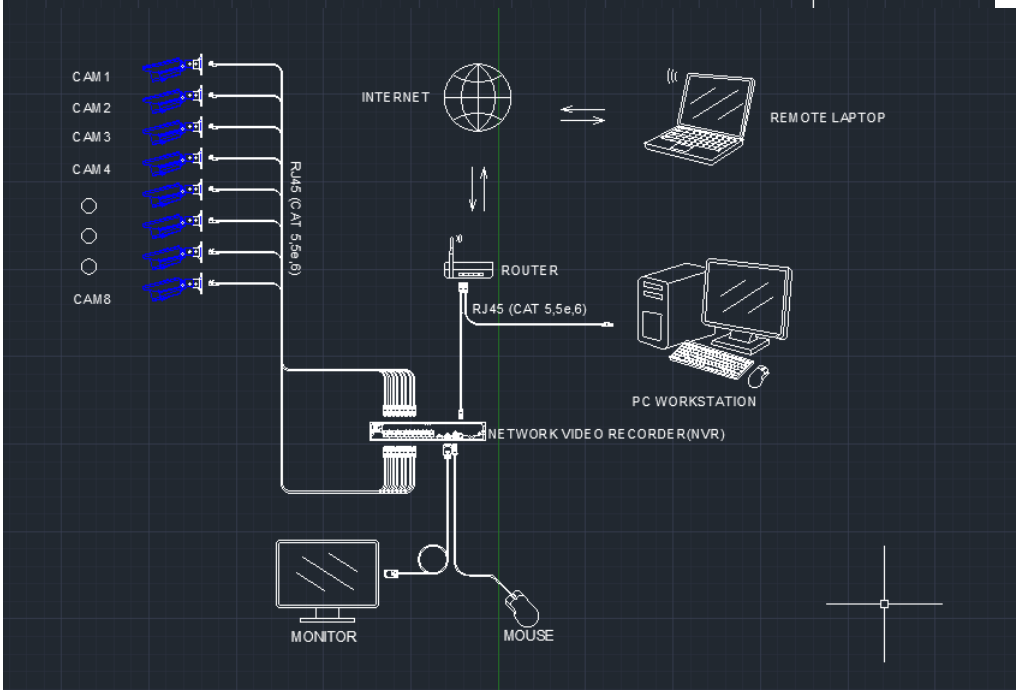


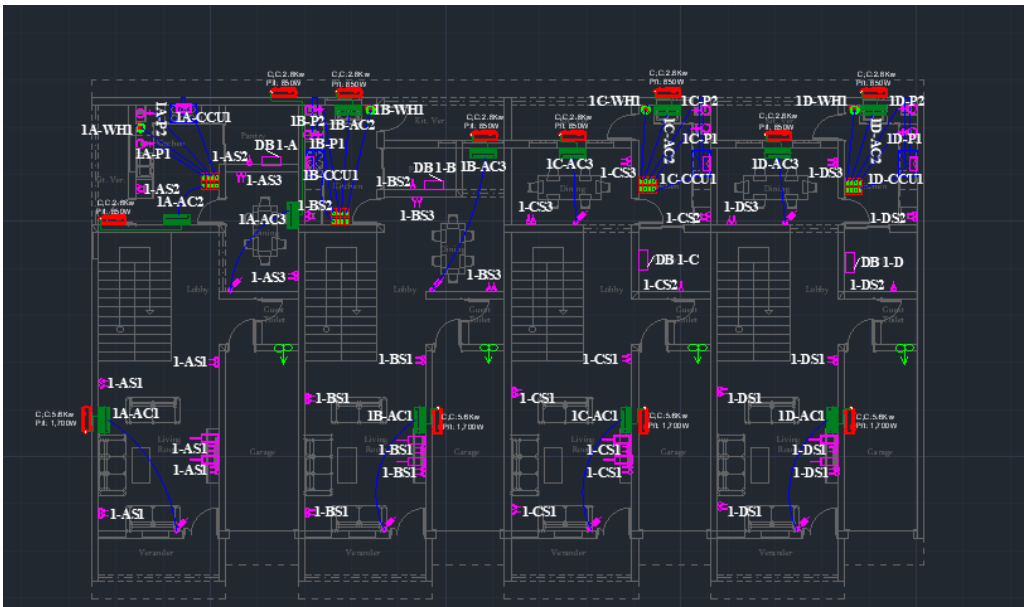
FIRST FLOOR PLAN


CAM7
CAM6
CAM8
CAM5
HD2
HD1
B2
MCP2
HD4
B3
HD5
MCP3
SD6
SD8
MCP4
B4
HD7
B1
MCP1
HD3
SD3
SD4
HD6
HD8
SD5
SD7
SD2
HD12
HD11
SD10
HD10
SD9
HD9
SD11
SD1
B8
MCP12
MCP11
FARP1
B7
MCP10
MCP9
FARP2
B6
MCP8
MCP7
FARP3
B5
MCP6
MCP5
CAM4
FACP
CAM1
CAM2
CAM3
GROUND FLOOR PLAN

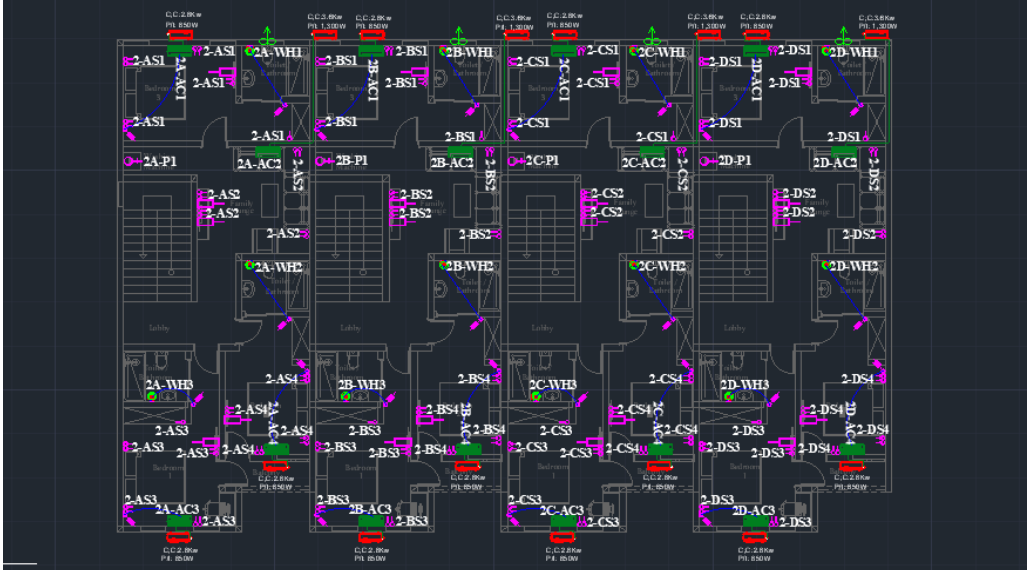

SD14
SD17
SD20
SD23
SD13
SD16
SD19
SD22
SD15
SD18
SD21
SD24
B9
MCP13
B12
MCP16
B11
MCP15
B10
MCP14
SD12
SD32
SD29
SD25
SD34
SD31
SD28
SD26
SD35
SD33
SD30
SD27
FIRST FLOOR PLAN


Fire Detection & Alarm Loop
1 x 2 x 1.5mm²
LOOP 1 GROUND FLOOR
FACP
1 Loop Addressable
Fire Alarm Panel
FARP1
FARP2
FARP3


CAM 1
CAM 2
CAM 3
CAM 4
CAM 8
RJ45 (CAT 5,5e,6)
INTERNET
REMOTE LAPTOP
ROUTER
RJ45 (CAT 5,5e,6)
PC WORKSTATION
NETWORK VIDEO RECORDER(NVR)
MONITOR
MOUSE



GROUND FLOOR PLAN



FIRST FLOOR PLAN

GROUND FLOOR DISTRIBUTION BOARD (DB 1-A)

Location: PANTRY
Voltage & Freq.: 230/400V, 50Hz
Cable Length(m) to ControlPanel
Type: 100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
DB Ref.: DB 1-A
Fed from CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
1A-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80	1360.00			1 2	1200			0.80	1500	1500	1	3 x 4.0	1A-WH1
1A-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		3 4		680.00		0.80	850	850	1	3 x 4.0	1A-AC2 - 20A SWITCH OUTLET
1A-P2 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40			800.00	5 6			1200.00	0.40	3000	3000	1	3 x 6.0	1A-CCU1
1-AL1 - LIGHTING POINTS	2 x 1.5	18	100	1800	0.45	810.00			7 8	0			0.40	0	2000	0	3 x 4.0	SPARE
1A-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		9 10		225		0.45	500	250	2	3 x 2.5	1-AS2 - SOCKET POINTS
1-AL2 - LIGHTING POINTS	2 x 1.5	18	100	1800	0.45			810.00	11 12			450.00	0.45	1000	1000	1	2 x 1.5	1-AL3 - LIGHTING POINTS
SPARE	3 x 4.0	0	2000	0	0.40	0.00			13 14	0			0.80	0	1500	0	3 x 4.0	SPARE
1-AS1 - SOCKET POINTS	3 x 2.5	5	250	1250	0.45		562.5		15 16		225		0.45	500	250	2	3 x 2.5	1-AS3 - SOCKET POINTS
SPARE	3 x 4.0	0	1200	0	1.00			0	17 18			0	1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.50	0			19 20	0			0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.80		0		21 22		0		0.45	0	250	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.45			0	23 24			0	0.70	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0			25 26	0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.60		0		27 28		0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0	29 30			0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0			31 32	0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0		33 34		0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0	35 36			0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				11400		2170	2042.5	1610		1200.00	1130	1650.00		7350				
* RING MAIN FINBL SUB-CIRCUITS									100A MCB; 300mA									

DB LOAD CABLE SIZE:
4C x 6mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
100A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	3370.00	3173	3260.00		
SPARE(W)				500	
TOTAL CONNECTED (W)				10302.50	W
POWER FACTOR				0.8	Cos ϕ
TOTAL CONNECTED (VA)				12878.125	VA
CURRENT (A)				18	A

EARTH WIRE SIZE:
1 x 6mm² PVC SINGLE CORE COPPER CABLE

FIRST FLOOR DISTRIBUTION BOARD (DB 2-A)

Location: LOBBY
Voltage & Freq.: 230/400V, 50Hz
Cable Length(m) to ControlPanel
Type: 100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
DB Ref.: DB 2-A
Fed from CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
2A-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80	680.00			1 2	562.5			0.45	1250	250	5	3 x 2.5	2-AS1 - SOCKET POINTS
2A-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		3 4		1040.00		0.80	1300	1300	1	3 x 4.0	2A-AC2 - 20A SWITCH OUTLET
2-AS2 - SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	5 6			1200.00	0.80	1500	1500	1	3 x 4.0	2A-WH1
2A-WH2	3 x 4.0	1	1500	1500	0.80	1200.00			7 8	680			0.80	850	850	1	3 x 4.0	2A-AC4 - 20A SWITCH OUTLET
2A-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		9 10		562.5		0.45	1250	250	5	3 x 2.5	2-AS3 - SOCKET POINTS
2-AL2 - LIGHTING POINTS	2 x 1.5	10	100	1000	0.45			450.00	11 12			1200.00	0.80	1500	1500	1	3 x 4.0	2A-WH3
2-AL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45	450.00			13 14	315			0.45	700	100	7	2 x 1.5	2-AL5 - LIGHTING POINTS
2-AL1 - LIGHTING POINTS	2 x 1.5	9	100	900	0.45		405		15 16		225		0.45	500	100	5	2 x 1.5	2-AL4 - LIGHTING POINTS
SPARE	3 x 4.0	0	1200	0	1.00			0	17 18			450	0.45	1000	250	4	3 x 2.5	2-AS4 - SOCKET POINTS
SPARE	3 x 2.5	0	250	0	0.50	0			19 20	0			0.45	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.80		0		21 22		0		0.40	0	6000	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.45			0	23 24			0	0.70	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0			25 26	0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.60		0		27 28		0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0	29 30			0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0			31 32	0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0		33 34		0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0	35 36			0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				9100		2330	1885	900		1557.50	1828	2650.00		9850				
* RING MAIN FINBL SUB-CIRCUITS									100A MCB; 300mA									

DB LOAD CABLE SIZE:
4C x 6mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
100A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	3887.50	3713	3750.00		
SPARE(W)				500	
TOTAL CONNECTED (W)				11850.00	W
POWER FACTOR				0.8	Cos ϕ
TOTAL CONNECTED (VA)				14812.5	VA
CURRENT (A)				20.7	A

EARTH WIRE SIZE:
1 x 6mm² PVC SINGLE CORE COPPER CABLE

GROUND FLOOR DISTRIBUTION BOARD (DB 1-B)

Location: PANTRY
Voltage & Freq.: 230/400V, 50Hz
Cable Length(m) to ControlPanel
Type: 100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
DB Ref.: DB 1-B
Fed from CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B	TRIP RATING	PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
1B-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80	1360.00			1 2	1200			0.80	1500	1500	1	3 x 4.0	1B-WH1
1B-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		3 4		562.50		0.45	1250	250	5	3 x 2.5	1-BS1 - SOCKET POINTS
1B-P2 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40			800.00	5 6			1200.00	0.40	3000	3000	1	3 x 6.0	1B-CCU1
SPARE	3 x 4.0	0	850	0	0.80	0.00			7 8	0			0.40	0	2000	0	3 x 4.0	SPARE
1B-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		9 10		225		0.45	500	250	2	3 x 2.5	1-BS2 - SOCKET POINTS
1-BL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45			450.00	11 12			680.00	0.80	850	850	1	3 x 4.0	1B-AC2 - 20A SWITCH OUTLET
1-BL1 - LIGHTING POINTS	2 x 1.5	13	100	1300	0.45	585.00			13 14	0			0.80	0	1500	0	3 x 4.0	SPARE
1-BL2 - LIGHTING POINTS	2 x 1.5	18	100	1800	0.45		720		15 16		225		0.45	500	250	2	3 x 2.5	1-BS3 - SOCKET POINTS
SPARE	3 x 4.0	0	1200	0	1.00			0	17 18			0	1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.50	0			19 20	0			0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.80		0		21 22		0		0.45	0	250	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.45			0	23 24			0	0.70	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0			25 26	0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.60		0		27 28		0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0	29 30			0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0			31 32	0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0		33 34		0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0	35 36			0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				10450		1945	2200	1250		1200.00	1013	1880.00		7600				
* RING MAIN FINBL SUB-CIRCUITS									100A MCB; 300mA									

DB LOAD CABLE SIZE:
4C x 6mm² PVC/PVC COPPER CABLE

DB ISOLATOR TYPE:
100A TP&N MCCB; 300mA

	R	Y	B		
CONNECTED (W)	3145.00	3213	3130.00		
SPARE(W)				500	
TOTAL CONNECTED (W)				9987.50	W
POWER FACTOR				0.8	Cos ϕ
TOTAL CONNECTED (VA)				12484.375	VA
CURRENT (A)				17.4	A

EARTH WIRE SIZE:
1 x 6mm² PVC SINGLE CORE COPPER CABLE

FIRST FLOOR DISTRIBUTION BOARD (DB 2-A)

Location:	LOBBY	**Type:**	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB 2-A
Cable Length(m) to ControlPanel		**Fed from**	CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B		TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
2B-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80	680.00			1		10	562.5			0.45	1250	250	5	3 x 2.5	2-BS1 - SOCKET POINTS
2B-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		2		11		1040.00		0.80	1300	1300	1	3 x 4.0	2B-AC2 - 20A SWITCH OUTLET
2-BS2 - SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	3		12			1200.00	0.80	1500	1500	1	3 x 4.0	2B-WH1
2B-WH2	3 x 4.0	1	1500	1500	0.80	1200.00			4		13	680			0.80	850	850	1	3 x 4.0	2B-AC4 - 20A SWITCH OUTLET
2B-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		5		14		562.5		0.45	1250	250	5	3 x 2.5	2-BS3 - SOCKET POINTS
2-BL2 - LIGHTING POINTS	2 x 1.5	10	100	1000	0.45			450.00	6		15			1200.00	0.80	1500	1500	1	3 x 4.0	2B-WH3
2-BL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45	450.00			7		16	315			0.45	700	100	7	2 x 1.5	2-BL5 - LIGHTING POINTS
2-BL1 - LIGHTING POINTS	2 x 1.5	9	100	900	0.45		405		8		17		225		0.45	500	100	5	2 x 1.5	2-BL4 - LIGHTING POINTS
SPARE	3 x 4.0	0	1200	0	1.00			0	9		18			450	0.45	1000	250	4	3 x 2.5	2-BS4 - SOCKET POINTS
SPARE	3 x 2.5	0	250	0	0.50	0						0			0.45	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.80		0						0		0.40	0	6000	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.45			0						0	0.70	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0						0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.60		0						0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0						0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0						0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0						0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0						0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				9100		2330	1885	900				1557.50	1828	2850.00		9850				
* RING MAIN FINBL SUB-CIRCUITS										100A MCB, 300mA										

DB LOAD CABLE SIZE:		R	Y	B		*EARTH WIRE SIZE:*
4C x 6mm^2 PVC/PVC COPPER CABLE	CONNECTED (W)	**3887.50**	**3713**	**3750.00**		**1 x 6mm^2 PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***11850.00***	W
100A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**14812.5**	VA
	CURRENT (A)				**20.7**	A

GROUND FLOOR DISTRIBUTION BOARD (DB 1-C)

Location:	PANTRY	**Type:**	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB 1-C
Cable Length(m) to ControlPanel		**Fed from**	CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B		TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
1C-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80	1360.00			1		10	1200			0.80	1500	1500	1	3 x 4.0	1C-WH1
1C-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		2		11		680.00		0.80	850	850	1	3 x 4.0	1C-AC2 - 20A SWITCH OUTLET
1C-P2 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40			800.00	3		12			1200.00	0.40	3000	3000	1	3 x 6.0	1C-CCU1
SPARE	3 x 4.0	0	850	0	0.80	0.00			4		13	0			0.40	0	2000	0	3 x 4.0	SPARE
1C-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		5		14		225		0.45	500	250	2	3 x 2.5	1-BS2 - SOCKET POINTS
1-CL2 - LIGHTING POINTS	2 x 1.5	14	100	1400	0.45			630.00	6		15			562.50	0.45	1250	250	5	3 x 2.5	1-BS1 - SOCKET POINTS
1-CL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45	450.00			7		16	0			0.80	0	1500	0	3 x 4.0	SPARE
1-CL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45		675		8		17		225		0.45	500	250	2	3 x 2.5	1-BS3 - SOCKET POINTS
SPARE	3 x 4.0	0	1200	0	1.00			0	9		18			0	1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 2.5	0	250	0	0.50	0						0			0.50	0	250	0	3 x 2.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.80		0						0		0.40	0	6000	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.45			0						0	0.70	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0						0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.60		0						0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0						0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0						0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0						0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0						0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				10450		1810	2155	1430				1200.00	1130	1762.50		7600				
* RING MAIN FINBL SUB-CIRCUITS										100A MCB, 300mA										

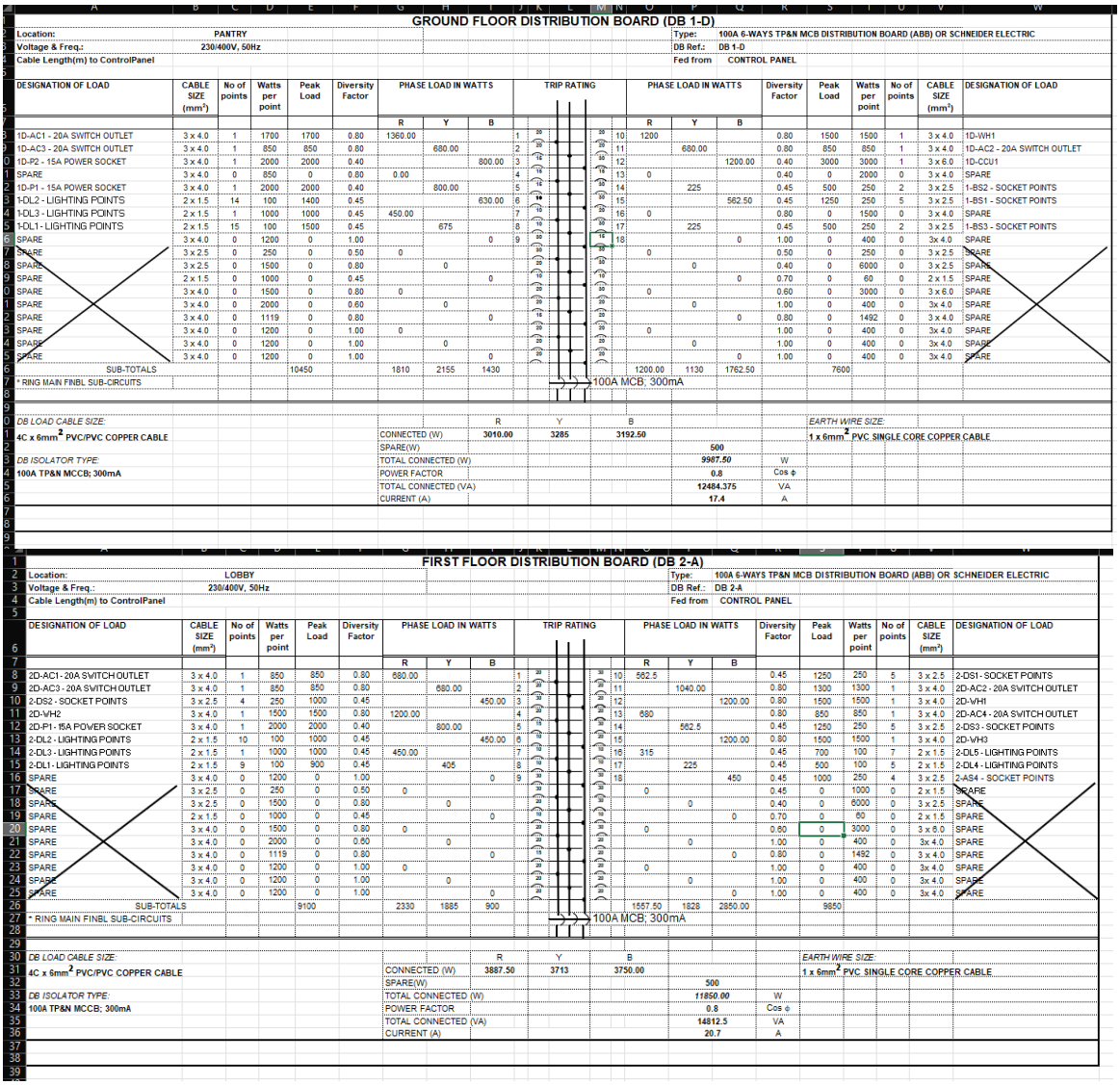
DB LOAD CABLE SIZE:		R	Y	B		*EARTH WIRE SIZE:*
4C x 6mm^2 PVC/PVC COPPER CABLE	CONNECTED (W)	**3010.00**	**3285**	**3192.50**		**1 x 6mm^2 PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***9987.50***	W
100A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**12484.375**	VA
	CURRENT (A)				**17.4**	A

FIRST FLOOR DISTRIBUTION BOARD (DB 2-A)

Location:	LOBBY	**Type:**	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC
Voltage & Freq.:	230/400V, 50Hz	**DB Ref.:**	DB 2-A
Cable Length(m) to ControlPanel		**Fed from**	CONTROL PANEL

DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS R	Y	B		TRIP RATING		PHASE LOAD IN WATTS R	Y	B	Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD
2C-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80	680.00			1		10	562.5			0.45	1250	250	5	3 x 2.5	2-CS1 - SOCKET POINTS
2C-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		2		11		1040.00		0.80	1300	1300	1	3 x 4.0	2C-AC2 - 20A SWITCH OUTLET
2-CS2 - SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	3		12			1200.00	0.80	1500	1500	1	3 x 4.0	2C-WH1
2C-WH2	3 x 4.0	1	1500	1500	0.80	1200.00			4		13	680			0.80	850	850	1	3 x 4.0	2C-AC4 - 20A SWITCH OUTLET
2C-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		5		14		562.5		0.45	1250	250	5	3 x 2.5	2-CS3 - SOCKET POINTS
2-CL2 - LIGHTING POINTS	2 x 1.5	10	100	1000	0.45			450.00	6		15			1200.00	0.80	1500	1500	1	3 x 4.0	2C-WH3
2-CL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45	450.00			7		16	315			0.45	700	100	7	2 x 1.5	2-CL5 - LIGHTING POINTS
2-CL1 - LIGHTING POINTS	2 x 1.5	9	100	900	0.45		405		8		17		225		0.45	500	100	5	2 x 1.5	2-CL4 - LIGHTING POINTS
SPARE	3 x 4.0	0	1200	0	1.00			0	9		18			450	0.45	1000	250	4	3 x 2.5	2-CS4 - SOCKET POINTS
SPARE	3 x 2.5	0	250	0	0.50	0						0			0.45	0	1000	0	2 x 1.5	SPARE
SPARE	3 x 2.5	0	1500	0	0.80		0						0		0.40	0	6000	0	3 x 2.5	SPARE
SPARE	2 x 1.5	0	1000	0	0.45			0						0	0.70	0	60	0	2 x 1.5	SPARE
SPARE	3 x 4.0	0	1500	0	0.80	0						0			0.60	0	3000	0	3 x 6.0	SPARE
SPARE	3 x 4.0	0	2000	0	0.60		0						0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1119	0	0.80			0						0	0.80	0	1492	0	3 x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00	0						0			1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00		0						0		1.00	0	400	0	3x 4.0	SPARE
SPARE	3 x 4.0	0	1200	0	1.00			0						0	1.00	0	400	0	3x 4.0	SPARE
SUB-TOTALS				9100		2330	1885	900				1557.50	1828	2850.00		9850				
* RING MAIN FINBL SUB-CIRCUITS										100A MCB, 300mA										

DB LOAD CABLE SIZE:		R	Y	B		*EARTH WIRE SIZE:*
4C x 6mm^2 PVC/PVC COPPER CABLE	CONNECTED (W)	**3887.50**	**3713**	**3750.00**		**1 x 6mm^2 PVC SINGLE CORE COPPER CABLE**
	SPARE(W)				**500**	
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)				***11850.00***	W
100A TP&N MCCB; 300mA	POWER FACTOR				**0.8**	Cos ϕ
	TOTAL CONNECTED (VA)				**14812.5**	VA
	CURRENT (A)				**20.7**	A



GROUND FLOOR DISTRIBUTION BOARD (DB 1-D)

Location:	PANTRY												Type:	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC										
Voltage & Freq.:	230/400V, 50Hz												DB Ref.:	DB 1-D										
Cable Length(m) to ControlPanel													Fed from	CONTROL PANEL										
DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS			TRIP RATING			PHASE LOAD IN WATTS			Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD					
						R	Y	B				R	Y	B											
1D-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	1700	1700	0.80	1360.00			1	2		10	1200			0.80	1500	1500	1	3 x 4.0	1D-WH1				
1D-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		3	4		11		680.00		0.80	850	850	1	3 x 4.0	1D-AC2 - 20A SWITCH OUTLET				
1D-P2 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40			800.00	5	6		12			1200.00	0.40	3000	3000	1	3 x 6.0	1D-CCU1				
SPARE	3 x 4.0	0	850	0	0.80	0.00			7	8		13	0			0.40	0	2000	0	3 x 4.0	SPARE				
1D-P1 - 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		9	10		14		225		0.45	500	250	2	3 x 2.5	1-BS2 - SOCKET POINTS				
1-DL2 - LIGHTING POINTS	2 x 1.5	14	100	1400	0.45			630.00	11	12		15			562.50	0.45	1250	250	5	3 x 2.5	1-BS1 - SOCKET POINTS				
1-DL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45	450.00			13	14		16	0			0.80	0	1500	0	3 x 4.0	SPARE				
1-DL1 - LIGHTING POINTS	2 x 1.5	15	100	1500	0.45		675		15	16		17		225		0.45	500	250	2	3 x 2.5	1-BS3 - SOCKET POINTS				
SPARE	3 x 4.0	0	1200	0	1.00			0	17	18		18			0	1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 2.5	0	250	0	0.50	0			19	20			0			0.50	0	250	0	3 x 2.5	SPARE				
SPARE	3 x 2.5	0	1500	0	0.80		0		21	22				0		0.40	0	6000	0	3 x 2.5	SPARE				
SPARE	2 x 1.5	0	1000	0	0.45			0	23	24					0	0.70	0	60	0	2 x 1.5	SPARE				
SPARE	3 x 4.0	0	1500	0	0.80	0			25	26			0			0.60	0	3000	0	3 x 6.0	SPARE				
SPARE	3 x 4.0	0	2000	0	0.60		0		27	28				0		1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 4.0	0	1119	0	0.80			0	29	30					0	0.80	0	1492	0	3 x 4.0	SPARE				
SPARE	3 x 4.0	0	1200	0	1.00	0			31	32			0			1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 4.0	0	1200	0	1.00		0		33	34				0		1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 4.0	0	1200	0	1.00			0	35	36					0	1.00	0	400	0	3x 4.0	SPARE				
SUB-TOTALS				10450		1810	2155	1430					1200.00	1130	1762.50		7600								
* RING MAIN FINBL SUB-CIRCUITS											100A MCB; 300mA														

DB LOAD CABLE SIZE:		R	Y	B
4C x 6mm² PVC/PVC COPPER CABLE	CONNECTED (W)	3010.00	3285	3192.50
	SPARE(W)			500
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)			9987.50 W
100A TP&N MCCB; 300mA	POWER FACTOR			0.8 Cos ϕ
	TOTAL CONNECTED (VA)			12484.375 VA
	CURRENT (A)			17.4 A

EARTH WIRE SIZE: 1 x 6mm² PVC SINGLE CORE COPPER CABLE

FIRST FLOOR DISTRIBUTION BOARD (DB 2-A)

Location:	LOBBY												Type:	100A 6-WAYS TP&N MCB DISTRIBUTION BOARD (ABB) OR SCHNEIDER ELECTRIC										
Voltage & Freq.:	230/400V, 50Hz												DB Ref.:	DB 2-A										
Cable Length(m) to ControlPanel													Fed from	CONTROL PANEL										
DESIGNATION OF LOAD	CABLE SIZE (mm^2)	No of points	Watts per point	Peak Load	Diversity Factor	PHASE LOAD IN WATTS			TRIP RATING			PHASE LOAD IN WATTS			Diversity Factor	Peak Load	Watts per point	No of points	CABLE SIZE (mm^2)	DESIGNATION OF LOAD					
						R	Y	B				R	Y	B											
2D-AC1 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80	680.00			1	2		10	562.5			0.45	1250	250	5	3 x 2.5	2-DS1 - SOCKET POINTS				
2D-AC3 - 20A SWITCH OUTLET	3 x 4.0	1	850	850	0.80		680.00		3	4		11		1040.00		0.80	1300	1300	1	3 x 4.0	2D-AC2 - 20A SWITCH OUTLET				
2-DS2 - SOCKET POINTS	3 x 2.5	4	250	1000	0.45			450.00	5	6		12			1200.00	0.80	1500	1500	1	3 x 4.0	2D-VH1				
2D-VH2	3 x 4.0	1	1500	1500	0.80	1200.00			7	8		13	680			0.80	850	850	1	3 x 4.0	2D-AC4 - 20A SWITCH OUTLET				
2D-P1- 15A POWER SOCKET	3 x 4.0	1	2000	2000	0.40		800.00		9	10		14		562.5		0.45	1250	250	5	3 x 2.5	2-DS3 - SOCKET POINTS				
2-DL2 - LIGHTING POINTS	2 x 1.5	10	100	1000	0.45			450.00	11	12		15			1200.00	0.80	1500	1500	1	3 x 4.0	2D-VH3				
2-DL3 - LIGHTING POINTS	2 x 1.5	1	1000	1000	0.45	450.00			13	14		16	315			0.45	700	100	7	2 x 1.5	2-DL5 - LIGHTING POINTS				
2-DL1 - LIGHTING POINTS	2 x 1.5	9	100	900	0.45		405		15	16		17		225		0.45	500	100	5	2 x 1.5	2-DL4 - LIGHTING POINTS				
SPARE	3 x 4.0	0	1200	0	1.00			0	17	18		18			450	0.45	1000	250	4	3 x 2.5	2-AS4 - SOCKET POINTS				
SPARE	3 x 2.5	0	250	0	0.50	0			19	20			0			0.45	0	1000	0	2 x 1.5	SPARE				
SPARE	3 x 2.5	0	1500	0	0.80		0		21	22				0		0.40	0	6000	0	3 x 2.5	SPARE				
SPARE	2 x 1.5	0	1000	0	0.45			0	23	24					0	0.70	0	60	0	2 x 1.5	SPARE				
SPARE	3 x 4.0	0	1500	0	0.80	0			25	26			0			0.60	0	3000	0	3 x 6.0	SPARE				
SPARE	3 x 4.0	0	2000	0	0.60		0		27	28				0		1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 4.0	0	1119	0	0.80			0	29	30					0	0.80	0	1492	0	3 x 4.0	SPARE				
SPARE	3 x 4.0	0	1200	0	1.00	0			31	32			0			1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 4.0	0	1200	0	1.00		0		33	34				0		1.00	0	400	0	3x 4.0	SPARE				
SPARE	3 x 4.0	0	1200	0	1.00			0	35	36					0	1.00	0	400	0	3x 4.0	SPARE				
SUB-TOTALS				9100		2330	1885	900					1557.50	1828	2850.00		9850								
* RING MAIN FINBL SUB-CIRCUITS											100A MCB; 300mA														

DB LOAD CABLE SIZE:		R	Y	B
4C x 6mm² PVC/PVC COPPER CABLE	CONNECTED (W)	3887.50	3713	3750.00
	SPARE(W)			500
DB ISOLATOR TYPE:	TOTAL CONNECTED (W)			11850.00 W
100A TP&N MCCB; 300mA	POWER FACTOR			0.8 Cos ϕ
	TOTAL CONNECTED (VA)			14812.5 VA
	CURRENT (A)			20.7 A

EARTH WIRE SIZE: 1 x 6mm² PVC SINGLE CORE COPPER CABLE

Asset Metadata

Item	Description
Project ID	P006
Building Type	Residential Terrace Building
Engineering Discipline	Electrical Engineering
Asset Category	Lighting, Power & ELV Systems
Software	AutoCAD, Microsoft Excel

File Formats	DWG, XLSX
Drawing Status	Completed Design

Project Overview

This project presents the electrical design documentation for a residential terrace building, integrating lighting, power distribution, fire alarm systems, CCTV layouts, and electrical load schedules into a coordinated engineering package suitable for residential installation.

Scope of Work

- Lighting layout design
- Power layout design
- Fire alarm layout preparation
- CCTV layout design
- Load calculation
- Load schedule preparation
- Circuit distribution planning

Engineering Assets Produced

- Lighting Layout Plans
- Power Layout Plans
- Fire Alarm Layout Plans
- CCTV Layout Plans
- Electrical Load Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout Plans	Lighting for TERRACE BUILDING	DWG
P-001	Power Layout Plans	Power for TERRACE BUILDING	DWG

SCH-001	Fire Alarm And CCTV Schematics Layout	ELV for TERRACE BUILDING	DWG
LS-001	Load Schedule	TERRACE BUILDING DB	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Microsoft Excel Load Schedule (.xlsx)

Design Considerations

The lighting design was developed to provide appropriate illumination levels and balanced circuit grouping across the building.

The power distribution system was planned to achieve safe socket outlet distribution, balanced circuit loading, appropriate protective device selection, and organised cable routing.

The fire alarm and CCTV systems were coordinated to ensure effective detector placement, surveillance coverage, maintainable cable routing, and seamless integration with the overall electrical installation.

Electrical load calculations established connected load, estimated demand, protective device selection criteria, cable sizing basis, and circuit balancing requirements.

Repository Structure

P006_Terrace_Building/

├── CAD_Drawings

├── README.md

└── Load_Schedules

Skills Demonstrated

- Residential electrical design
- Lighting and power distribution planning
- ELV system coordination
- Load calculation and schedule preparation
- Circuit distribution planning
- Professional engineering documentation

PROJECT SUMMARY

Above are selected engineering projects carefully done in accordance with engineering standards. I believe they would be of great help in improving AI systems.